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CRYOGENIC MAGNETIC FIELD MONITORING SYSTEM IN THE SNS NEUTRON EDM EXPERIMENT

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CRYOGENIC MAGNETIC FIELD MONITORING SYSTEM IN THE SNS
NEUTRON EDM EXPERIMENT

DISSERTATION

A dissertation submitted in partial
fulfillment of the requirements for
the degree of Doctor of Philosophy
in the College of Arts and Sciences
at the University of Kentucky

By
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Lexington, Kentucky

Director: Dr. Bradley Plaster, Professor of Physics
Lexington, Kentucky
2022

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ABSTRACT OF DISSERTATION

CRYOGENIC MAGNETIC FIELD MONITORING SYSTEM IN THE SNS NEUTRON EDM EXPERIMENT

A permanent neutron electric dipole moment (nEDM), d_n , would violate charge-conjugation and parity (CP) symmetry. The neutron electric dipole moment currently has a global limit of $d_n < 1.8 \times 10^{-26}$ e·cm (90% CL). This limit is intended to be improved by two orders of magnitude by the nEDM experiment at the Spallation Neutron Source (SNS nEDM), $d_n \sim 10^{-28}$ e·cm. The magnetic field non-uniformities within the experimental region must be precisely monitored and managed in order to suppress systematic effects in the experiment caused by magnetic field gradients. The estimation of magnetic field components within an experimentally inaccessible zone is a challenging problem to solve, particularly in situations where direct field measurements are not feasible. To tackle this issue in the SNS nEDM experiment, a magnetic field monitoring system consisting of 39 single-axis cryogenic fluxgate magnetometer probes at discrete locations has been designed and built to provide a first-pass measurement of the field gradients within the experimental region via nondisruptive measurements of the magnetic field's components. Room-temperature tests of the system were carried out at the University of Kentucky, followed by cryogenic tests of the system at the California Institute of Technology. This dissertation will describe the theoretical framework of the magnetic field reconstruction approach and present the results of the system's preliminary tests. In addition, we propose a new method of accessing magnetic field components using physics-informed neural networks (PINNs). The new method we present will be of broad interest to experiments requiring precise determination of magnetic field components.

KEYWORDS: neutrons, electric dipole moment, magnetic field, neural networks

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September 29, 2022

CRYOGENIC MAGNETIC FIELD MONITORING SYSTEM IN THE SNS
NEUTRON EDM EXPERIMENT

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Date

Dedicated to my family.

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Chapter 1 Introduction

It is now known that all matter in the Universe is composed of a number of fundamental particles that are governed by four fundamental forces discovered by thousands of physicists since the discovery of quantum nature of the Universe. In the Standard Model (SM) of particle physics, we are guided by our best understanding of the relationships between these particles and three forces. Its purpose is to explain the fundamental particles that are present in all matter as well as the forces that control how these particles interact with one another. It is the theory of particle physics that has had the most success to this point, and its predictions have been proven by a significant amount of experimental data. Quantum physics and the concept of symmetry form the basis of the Standard Model, which is built on these two major ideas. The Standard Model is able to provide an explanation for a diverse set of phenomena thanks to these guiding principles. These phenomena vary from the behavior of subatomic particles to the characteristics of atoms. The Standard Model is predicated on the presence of three primary forces: the electromagnetic force, the strong force, and the weak force. Different kinds of particles are responsible for the transmission of each of these forces. Photons are what mediate the electromagnetic force, gluons are what mediate the strong force, and $W^\pm$ and Z^0 bosons are what mediate the weak force. The nucleus of an atom is made up of protons and neutrons, which are made up of quarks. The nucleus is held together by the strong force, whereas radioactive decay is controlled by the weak force. While electromagnetism is in charge of determining how electrons orbit the nucleus, gravity is responsible for drawing things together.

While the Standard Model is able to explain a wide range of phenomena, it does not include gravity. Gravity is a force that is described by the theory of general relativity, which is not compatible with the Standard Model. Because of this, physicists

have been trying to come up with a theory that accounts for gravity along with the other natural forces. As a standard model extension, supersymmetry has promise. The theory of supersymmetry proposes that each of the particles described by the Standard Model must also be accompanied by a superpartner particle. The mass of these superpartner particles is identical to that of their Standard Model counterparts, but their spins are distinct. If the theory of supersymmetry is right, it might assist to explain the hierarchy problem. The hierarchy problem refers to the mystery surrounding the apparent disparity in strength between gravity and the other fundamental forces. It's likely that some of the contributions from quantum processes that would ordinarily make gravity stronger are cancelled out by supersymmetry, making that explanation one of the probable explanations for this phenomenon [8]. Furthermore, the Standard Model does not account for the existence of dark matter, which is necessary to explain galaxy rotation curves and the anisotropies in the cosmic microwave background of galaxies. According to the Standard Model, the particles that are already understood to make up the Universe do not interact with the hypothetical particles that make up dark matter. This leads scientists to conclude that dark matter is composed of a previously unknown sort of particle.

Table 1.1: A list of the elementary particles of the Standard Model.

Fermions (Matter)				
	Quarks		Leptons	
Generation	$Q = -1/3$	$Q = +2/3$	$Q = -1$	$Q = 0$
First	down (d)	up (u)	electron (e)	electron neutrino (ν_e)
Second	strange (s)	charm (c)	muon (μ)	muon neutrino (ν_μ)
Third	bottom (b)	top (t)	tau (τ)	tau neutrino (ν_τ)
Bosons (Force carriers)				
Field	Electromagnetic	Strong	Weak	Higgs
Name	photon (γ)	gluon (g)	$W^\pm, Z^0$	H

1.1 Fundamental Discrete Symmetries and Their Violations

The three most important discrete symmetries are charge conjugation (**C**), parity (**P**), and time reversal (**T**). These symmetries are violated in weak interactions, which mediate the processes of beta decay, nuclear fission and fusion. The Standard Model (SM) of particle physics also includes the discrete symmetry of **CP**, which is a combination of **C** and **P**. This symmetry is also violated in weak interactions. Electric Dipole Moments (EDMs) violate **P** and **T** symmetries and according to the **CPT** theorem are **T** violation is equivalent to **CP** violation. A nonzero value of an EDM would imply that **CP** is not a good symmetry of nature, and would be a clear sign of physics beyond the Standard Model. A variety of new physics models, such as Supersymmetry, Grand Unified Theories (GUTs), and models with extra spatial dimensions, can give rise to EDMs. The **CPT** symmetry is the combination of these three discrete symmetries, and states that any physical process must be invariant under the simultaneous transformation of all three symmetries. This symmetry is believed to be exact in nature, and **CPT** violation has not been observed so far. A violation of **CPT** would imply a breakdown of Lorentz invariance, and would have far-reaching consequences for theoretical physics and our understanding of nature.

1.1.1 Parity

The parity operator, **P**, is the operator that describes a reflection about the origin. This means that the parity operator would flip the sign of the coordinates as

$$\mathbf{P} : \begin{pmatrix} x \\ y \\ z \end{pmatrix} \rightarrow \begin{pmatrix} -x \\ -y \\ -z \end{pmatrix}. \quad (1.1)$$

Since **P** is a unitary operator, we expect that the position operator should satisfy,

$$\mathbf{P}^\dagger \mathbf{r}_i \mathbf{P} = -\mathbf{r}_i, \quad (1.2)$$

and let $|\vec{r}'\rangle$ be an eigenstate of the position operator $\mathbf{r}_i$,

$$\mathbf{r}_i |\vec{r}'\rangle = r'_i |\vec{r}'\rangle. \quad (1.3)$$

where $\vec{r}' = (x', y', z')$ is a position vector in three dimensional space. Acting $\mathbf{P}$ on both sides of Eq. 1.3 and making use of Eq. 1.2 gives

$$\mathbf{r}_i (\mathbf{P} |\vec{r}'\rangle) = -r'_i (\mathbf{P} |\vec{r}'\rangle). \quad (1.4)$$

We can see that the state $\mathbf{P} |\vec{r}'\rangle$ is an eigenstate of the position operator $\mathbf{r}_i$ with eigenvalue $-r'_i$. Hence, we can say that

$$\mathbf{P} |\vec{r}'\rangle = e^{i\phi} |-\vec{r}'\rangle \quad (1.5)$$

where ϕ is a phase. Note that the phase factor $e^{i\phi}$ is not observable, so we can set the overall phase factor term to 1, $e^{i\phi} = 1$. Applying the parity operator twice restores the initial state,

$$\mathbf{P}^2 |\vec{r}'\rangle = |\vec{r}'\rangle. \quad (1.6)$$

Therefore, we can conclude that the eigenvalues of the parity operator $\mathbf{P}$ are $P = \pm 1$.

In Eq. 1.2 we see that $\mathbf{r}_i$ is odd under parity. Vectors with odd transformation properties under parity are called (polar) vectors. We can also see how the angular momentum operator, $\mathbf{L}_k = \epsilon_{ijk} \mathbf{r}_i \mathbf{p}_j$, behaves under parity,

$$\mathbf{P}^\dagger \mathbf{L}_k \mathbf{P} = \epsilon_{ijk} (-\mathbf{r}_i) (-\mathbf{p}_j) = \mathbf{L}_k. \quad (1.7)$$

where ϵ_{ijk} is the Levi-Civita symbol. Unlike the position operator, the angular momentum operator is even under parity. Vectors with even transformation properties under parity are called axial vectors (or pseudovectors). Scalar operators can also be odd (pseudo scalar) or even under parity. For example, $\mathbf{L} \cdot \mathbf{S}$ is a scalar and $\mathbf{S} \cdot \mathbf{r}$ is a pseudoscalar.

1.1.2 Change Conjugation

In physics, charge conjugation is an operation that takes a particle state, $|p\rangle$, and converts it into the corresponding antiparticle state, $|\bar{p}\rangle$. The charge conjugation operator (or **C**-parity) is usually denoted by the symbol **C**,

$$\mathbf{C} |p\rangle = |\bar{p}\rangle. \quad (1.8)$$

The charge conjugation operator flips the sign of all internal discrete quantum numbers, such as charm and lepton number while leaving the mass, the energy, the momentum and the spin untouched. Photons have odd **C**-parity, $\mathbf{C} |\gamma\rangle = -|\gamma\rangle$, since the electric charge changes sign and, therefore, the current changes direction under **C**. Eigenstates of **C** can only be neutral particles (e.g. J/ψ) and neutral composite systems (e.g. electron-positron pair).

1.1.3 Time Reversal

In physics, time reversal symmetry is a fundamental symmetry of nature described by the invariance of physical laws under a reversal of the direction of the flow of time. Time reversal symmetry means that the laws of physics are the same regardless of whether time is flowing forward or backward.

The canonical commutation relation in quantum mechanics is a fundamental relationship between the position and momentum operators of a particle. It states that the position and momentum operators do not commute, and is represented by the Eq.:

$$[\mathbf{r}_i, \mathbf{p}_j] = i\hbar\delta_{ij} \quad (1.9)$$

where $\mathbf{r}_i$ is the position operator, $\mathbf{p}_j$ is the momentum operator, and $\hbar$ is the reduced Planck constant. The canonical commutation relation is a fundamental result of quantum mechanics that has important implications for the behavior of particles at

the atomic and subatomic level. Under time reversal, the position and the momentum operators transform as

$$\mathbf{r}_i \rightarrow \mathbf{r}_i \quad \text{and} \quad \mathbf{p}_j \rightarrow -\mathbf{p}_j$$

Therefore, $[\mathbf{r}_i, \mathbf{p}_j] \rightarrow -i\hbar\delta_{ij}$. This is not the same as the canonical commutation relation, which means that the canonical commutation relation does not hold under time reversal. To remedy this issue, we can write down the time reversal operator in the Heisenberg picture as,

$$\begin{aligned} \mathbf{T}|\alpha\rangle &= |\alpha'\rangle, \\ \mathbf{T}\mathbf{O}(t)\mathbf{T}^\dagger &= \mathbf{O}'(t') \end{aligned} \tag{1.10}$$

where $|\alpha\rangle$ is a quantum state and $\mathbf{O}(t)$ is an observable. From classical mechanics we expect that,

$$\begin{aligned} \mathbf{T}\mathbf{r}_i\mathbf{T}^\dagger &= \mathbf{r}_i \\ \mathbf{T}\mathbf{p}_j\mathbf{T}^\dagger &= -\mathbf{p}_j. \end{aligned} \tag{1.11}$$

Transforming Eq. 1.9 under time reversal gives

$$\begin{aligned} \mathbf{T}[\mathbf{r}_i, \mathbf{p}_j]\mathbf{T}^\dagger &= \mathbf{T}i\hbar\delta_{ij}\mathbf{T}^\dagger \\ -[\mathbf{r}_i, \mathbf{p}_j] &= \mathbf{T}i\delta_{ij}\mathbf{T}^\dagger\hbar \\ -i &= \mathbf{T}i\mathbf{T}^\dagger. \end{aligned} \tag{1.12}$$

Therefore, we see that the time reversal operator must be an antiunitary operator to preserve the canonical commutation relation.

1.1.4 P Violation: Wu et al. Experiment

In 1956, two physicists working independently, T.D. Lee and C.N. Yang, proposed that the weak nuclear force, one of the four fundamental forces of nature, does not obey parity symmetry [29]. In other words, the weak force can tell the difference between a left and a right. The two physicists proposed an experiment to test their

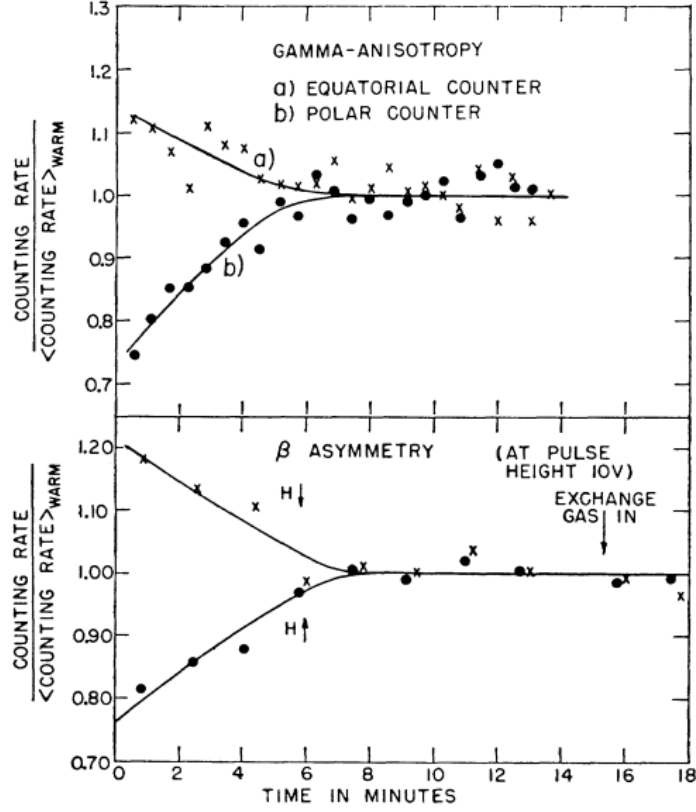
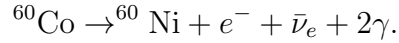


Figure 1.1: Results of the ^{60}Co beta decay experiment. Top: Gamma anisotropy between the polar and the equatorial directions. Bottom: Beta decay asymmetry in the direction parallel and antiparallel to the polarization of the ^{60}Co sample. Both plots show an anisotropy ≈ 0.6 . (Fig. from [48]).

theory, which was quickly carried out by C.S. Wu and her collaborators [48]. The experiment involved the decay of a ^{60}Co nucleus



A sample of ^{60}Co was cooled down and polarized by applying a high magnetic field in a particular direction. Two gamma scintillation counters were placed in the equatorial plane and close to polar position. To count the the beta particles emitted, another scintillation counter was placed at the polar position. Fig. 1.1 shows that the electrons are emitted preferentially in the opposite direction of the ^{60}Co spin. This difference could only be explained if parity was violated in the weak nuclear force. The $\mathbf{P}$ violation experiment was a groundbreaking discovery that confirmed the existence of

a previously unknown asymmetry in the laws of physics. It also opened the door to a new understanding of the weak nuclear force and its role in the Universe. Although $\mathbf{P}$ is violated, it was thought that combination of $\mathbf{C}$ and $\mathbf{P}$ symmetries, shortly $\mathbf{CP}$, was conserved.

1.1.5 CP Violation: Neutral Kaon Decay

Mesons composed of a (anti-)strange and up or down quark are called kaons. Kaons can be charged $K^\pm$ or neutral K^0 ($\bar{K}^0$). Strong interactions are necessary for the construction of kaons, while weak interactions are responsible for their decay. Since the strangeness is not conserved in the weak interactions, K^0 and its antiparticle $\bar{K}^0$ can transform into each other. This phenomena is known as neutral kaon mixing. The dynamics of this mixing can be described by the following Schrodinger equation [45, 10]:

$$i\hbar \frac{d}{dt} |\psi\rangle = \mathbf{H}' |\psi\rangle \quad (1.13)$$

where $|\psi\rangle = a(t) |K^0\rangle + b(t) |\bar{K}^0\rangle$ and $\mathbf{H}'$ is an effective Hamiltonian of the $K^0 - \bar{K}^0$ mixing. Also note that $\mathbf{H}'$ is non-Hermitian because of other kaon decays that are not contributing the mixing dynamics. $\mathbf{H}'$ can be decomposed into two Hermitian matrices as

$$\mathbf{H}' = \mathbf{M} - \frac{i}{2}\mathbf{\Gamma} = \begin{pmatrix} M_{11} & M_{12} \\ M_{12}^* & M_{11} \end{pmatrix} - \frac{i}{2} \begin{pmatrix} \Gamma_{11} & \Gamma_{12} \\ \Gamma_{12}^* & \Gamma_{11} \end{pmatrix}. \quad (1.14)$$

So far we only assumed $\mathbf{CPT}$ symmetry that implies that $\mathbf{M}_{11} = \mathbf{M}_{22}$ and $\mathbf{\Gamma}_{11} = \mathbf{\Gamma}_{22}$. If $\mathbf{CP}$ were a good symmetry of the nature, that would imply $\text{Im}(M_{12}) = \text{Im}(\Gamma_{12}) = 0$. The eigenstates of $\mathbf{H}'$ are

$$\begin{aligned} |K_1\rangle &= (|p|^2 + |q|^2)^{-1/2} (p |K^0\rangle + q |\bar{K}^0\rangle), \\ |K_2\rangle &= (|p|^2 + |q|^2)^{-1/2} (p |K^0\rangle - q |\bar{K}^0\rangle), \end{aligned} \quad (1.15)$$

with

$$\frac{q}{p} = \sqrt{\frac{\mathbf{H}'_{21}}{\mathbf{H}'_{12}}}. \quad (1.16)$$

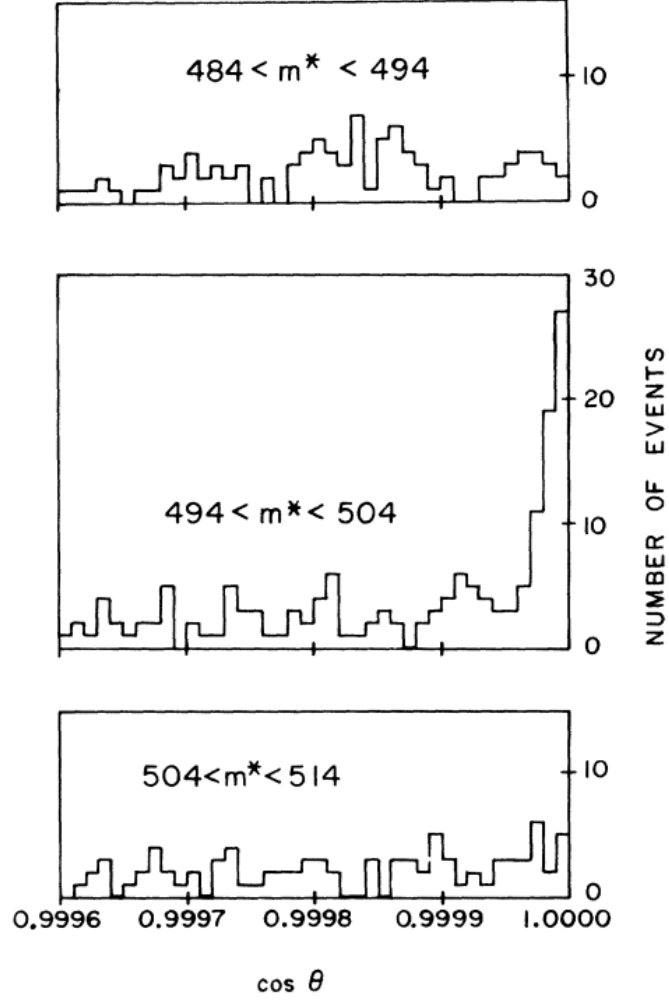


Figure 1.2: Histograms of angular distributions for $\pi^+\pi^-$ events. Center: There is a peak around $\cos(\theta) \approx 1$ in the neutral kaon mass range. (Fig. from [12]).

We stated that if **CP** were conserved, the off-diagonal elements of **H'** would be real ($q/p = 1$) and **CP** eigenstates would be

$$|K_1\rangle = \frac{|K^0\rangle - |\bar{K}^0\rangle}{\sqrt{2}}, \quad |K_2\rangle = \frac{|K^0\rangle + |\bar{K}^0\rangle}{\sqrt{2}}. \quad (1.17)$$

Experimentally, only two modes of neutral kaon decay were known: K_L^0 with mean lifetime $\tau_L = (5.116 \pm 0.021) \times 10^{-8}$ s and K_S^0 with mean lifetime $\tau_S = (8.954 \pm 0.004) \times 10^{-11}$ s. Neutral kaons mainly decay into two or three pions, e.g.,

$$\begin{aligned} K_S^0 &\rightarrow \pi^+ + \pi^- & \text{or} & & K_S^0 &\rightarrow \pi^0 + \pi^0 \\ K_L^0 &\rightarrow \pi^+ + \pi^- + \pi^0 & \text{or} & & K_L^0 &\rightarrow \pi^0 + \pi^0 + \pi^0. \end{aligned} \quad (1.18)$$

CP conserving eigenstates (see Eq. 1.17), $|K_1\rangle$, $|K_2\rangle$ are odd and even under **CP**, respectively. This strongly suggested that $|K_1\rangle = |K_S\rangle$ and $|K_2\rangle = |K_L\rangle$. In 1964, the first experimental evidence of **CP** violation was observed in neutral kaon decays [12]. In the experiment conducted by Christenson et. al., they had a beam of kaons traveling through a long tube, and the tube is long enough that the all short-lived kaons have time to decay. When the beam reaches the detector, it is composed only of long-lived kaons. However, they observed a small number of events where K_L decays into two pions (Fig. 1.2). This showed that $|K_S\rangle$ and $|K_L\rangle$ are not eigenstates of **CP** but their combinations are;

$$\begin{aligned} |K_S\rangle &= \frac{1}{\sqrt{1+\epsilon^2}}(|K_1\rangle - \epsilon |K_2\rangle) \\ |K_L\rangle &= \frac{1}{\sqrt{1+\epsilon^2}}(|K_2\rangle + \epsilon |K_1\rangle) \end{aligned} \quad (1.19)$$

with $\epsilon = 2 \times 10^{-3}$.

1.1.6 The Cabibbo-Kobayashi-Maskawa Matrix

The $W^\pm$ boson interactions with fermions are known as charged current interactions. Charged current interactions change the fermion flavour. In the Standard Model, the charged current Lagrangian is given as

$$\mathcal{L}_{CC} = -\frac{g}{\sqrt{2}}(\bar{U}_L \gamma^\mu W_\mu^+ \mathbf{V}_{CKM} D_L + \bar{D}_L \gamma^\mu W_\mu^- \mathbf{V}_{CKM}^\dagger U_L) \quad (1.20)$$

where $\mathbf{V}_{CKM}$ is called the Cabibbo-Kobayashi-Maskawa (CKM) matrix [25] which is a unitary matrix that encodes the strengths of the weak interactions between quarks of different generations:

$$\begin{pmatrix} d' \\ s' \\ b' \end{pmatrix} = \mathbf{V}_{CKM} \begin{pmatrix} d \\ s \\ b \end{pmatrix}. \quad (1.21)$$

Here the magnitude squared of the matrix element $(\mathbf{V}_{CKM})_{ij}$ denotes the probability of the transition from a i -type quark to j -type quark. There are several parametrizations

of the CKM matrix. A popular parameterization called the standard parametrization was introduced by Chau and Keung [11]. In this parametrization, the CKM matrix can be written explicitly as,

$$\mathbf{V}_{\text{CKM}} = \begin{pmatrix} c_{12}c_{13} & s_{12}c_{13} & s_{13}e^{-i\delta} \\ -s_{12}c_{23} - c_{12}s_{23}s_{13}e^{i\delta} & c_{12}c_{23} - s_{12}s_{23}s_{13}e^{i\delta} & s_{23}c_{13} \\ s_{12}s_{23} - c_{12}c_{23}s_{13}e^{i\delta} & -c_{12}s_{23} - s_{12}c_{23}s_{13}e^{i\delta} & c_{23}c_{13} \end{pmatrix} \quad (1.22)$$

where $s_{ij} = \sin(\theta_{ij})$ and $c_{ij} = \cos(\theta_{ij})$. As we can see, the CKM matrix can be determined by using only three rotation angles, θ_{12} , θ_{13} and θ_{23} , and a phase δ . Some terms in the CKM matrix (Eq. 1.22) are allowed to be complex and the charged-current Lagrangian (Eq. 1.20) is not invariant under **T** and **CP** in presence of complex terms. All of the CKM matrix elements, real and imaginary, must be measured in order to investigate the Standard Model's predictions. The most up-to-date values for the magnitudes of the CKM matrix elements are [47]

$$\mathbf{V}_{\text{CKM}} = \begin{pmatrix} 0.97401 \pm 0.00011 & 0.22650 \pm 0.00048 & 0.00361^{+0.00011}_{-0.00009} \\ 0.22636 \pm 0.00048 & 0.97320 \pm 0.00011 & 0.04053^{+0.00083}_{-0.00061} \\ 0.00854^{+0.00023}_{-0.00016} & 0.03978^{+0.00082}_{-0.00060} & 0.999172^{+0.000024}_{-0.000035} \end{pmatrix} \quad (1.23)$$

1.1.7 CP Violation in QCD

We have seen that **CP** is violated in weak interactions. It has been demonstrated that the strong interactions of the Standard Model, known as quantum chromo dynamics (QCD), have a non-trivial vacuum structure which makes it possible to violate **CP**. Symmetry and renormalization properties of the QCD Lagrangian allows a specific term which violates both parity and time reversal [10]:

$$\mathcal{L}_\theta = \frac{\theta g_s^2}{32\pi^2} \epsilon_{\mu\nu\alpha\beta} G_a^{\mu\nu} G^{a\alpha\beta} \quad (1.24)$$

where $G_{\mu\nu}^a$ is the gluon field strength tensor. The electric dipole moment of the neutron provides a particularly sensitive test of the **CP** conservation in QCD. The

term θ is one that violates **CP** and results in the neutron having an electric dipole moment, $d_n \approx 10^{-16}\theta$ e·cm [14]. The current world limit for the neutron electric dipole moment is $d_n = (0.0 \pm 1.1) \times 10^{-26}$ e·cm [2] which implies $\theta \lesssim 10^{-10}$. The Peccei–Quinn hypothesis, which was developed in 1977, hypothesized the existence of an elementary particle called an axion in an effort to overcome the strong-CP problem in QCD.

1.2 The Asymmetry between Matter and Antimatter

The observable Universe’s apparent imbalance between baryons and antibaryons is a sign of the asymmetry between matter and antimatter. This imbalance is also called the baryon asymmetry of the Universe (BAU). Generally, the BAU is measured in terms of the present Universe’s baryon-to-photon ratio, η :

$$\eta = \frac{n_B}{n_\gamma} = 6.1^{+0.3}_{-0.2} \times 10^{-10} \quad (1.25)$$

where n_B is the baryon number density and n_γ is the number density of photons.

The difficulty is that the events of the early Universe, such as the Big Bang and inflation, which are assumed to be responsible for the formation of matter, should have created equal amounts of matter and antimatter. On the other hand, the currently observable Universe is almost completely made up of matter, with only a trace quantity of antimatter present. The origin of this imbalance in the Universe remains a mystery.

The Sakharov conditions are a set of three necessary and sufficient conditions formulated by Andrei Sakharov in 1967 [40] for a process to create a net excess of particles over antiparticles in the early Universe. To induce such an imbalance, Sakharov identified the following requirements:

- Baryon number violation:

The quantity of baryonic matter that is contained within a particular body or

particle is referred to as its baryon number.

$$B = \frac{1}{3}(n_q - n_{\bar{q}}) \quad (1.26)$$

where n_q is the number of quarks and $n_{\bar{q}}$ is the number of antiquarks. The entire amount of baryonic matter in the Universe is always preserved by the Standard Model of particle physics. There must be an equivalent quantity of anti-baryon created or destroyed in each process that produces baryons.

- **C** and **CP** violation:

Suppose a **C**-symmetric process like

$$X \rightarrow Y + Z. \quad (1.27)$$

Then the assumption of **C**-symmetry requires that

$$\Gamma(X \rightarrow Y + Z) = \Gamma(\bar{X} \rightarrow \bar{Y} + \bar{Z}). \quad (1.28)$$

This gives no net baryon generation during the baryogenesis. Let us also consider the role of **CP**. Assume a particle X can decay into

$$\begin{aligned} X &\rightarrow q_L q_L \\ X &\rightarrow q_R q_R \end{aligned} \quad (1.29)$$

where q_L is a left-handed quark and q_R is a right-handed quark. **CP**-symmetry implies

$$\begin{aligned} \Gamma(X \rightarrow q_L q_L) &= \Gamma(\bar{X} \rightarrow \bar{q}_R \bar{q}_R) \\ \Gamma(X \rightarrow q_R q_R) &= \Gamma(\bar{X} \rightarrow \bar{q}_L \bar{q}_L) \\ \implies \Gamma(X \rightarrow q_L q_L) + \Gamma(X \rightarrow q_R q_R) &= \Gamma(\bar{X} \rightarrow \bar{q}_R \bar{q}_R) + \Gamma(\bar{X} \rightarrow \bar{q}_L \bar{q}_L) \end{aligned} \quad (1.30)$$

This again does not give any baryon asymmetry during the baryogenesis.

- Deviation from thermal equilibrium:

Assuming the **CPT** invariance, we can show that [46],

$$\begin{aligned}
\langle \mathbf{B} \rangle &= \text{Tr} [e^{\beta \mathbf{H}} \mathbf{B}] \\
&= \text{Tr} [\mathbf{CPT}(\mathbf{CPT})^{-1} e^{\beta \mathbf{H}} \mathbf{B}] \\
&= \text{Tr} [e^{\beta \mathbf{H}} (\mathbf{CPT}) \mathbf{B} (\mathbf{CPT})^{-1}] \\
&= -\text{Tr} [e^{\beta \mathbf{H}} \mathbf{B}]
\end{aligned} \tag{1.31}$$

where $\mathbf{B}$ is the baryon number operator, $\langle \cdot \rangle$ corresponds to the equilibrium average and $\beta = 1/T$. This implies that $\langle \mathbf{B} \rangle = 0$. From this we see that any opposite processes during the baryogenesis like

$$qq \rightarrow X$$

$$\bar{q}\bar{q} \rightarrow \bar{X}$$

will annihilate each other.

CP violation has been found in the decays of K , B and D mesons, but not at a magnitude that is sufficient to explain the BAU. As a result, to explain the matter-antimatter asymmetry of the Universe, we need to investigate other systems that violate **CP**, like the neutron electric dipole moment.

Chapter 2 Neutron Electric Dipole Moment

In chapter 1, we stated that **CP** violation in the Standard Model originates from two mechanisms: the QCD θ term and the complex phase δ in the CKM matrix. However, the **CP** mechanisms in the Standard Model are not able to explain the symmetry between matter and antimatter in the universe which clearly indicates that the Standard Model is definitely not the final theory of the universe. In order to solve this issue, physicist have proposed new theories beyond the Standard Model. One popular example is the Minimal Supersymmetric Standard Model (MSSM) [3]. The MSSM is the most straightforward supersymmetric extension of the Standard Model with the fewest additional free parameters possible. All of the Standard Model particles and their supersymmetric companions are included in the MSSM. Two Higgs doublets, two higgsinos, four gauginos, and six sfermions are among the new particles.

2.1 Neutron Electric Dipole Moment

The separation of positive and negative charges produces an electric dipole moment (see Fig. 2.1). The amplitude of the dipole moment depends on the magnitude of

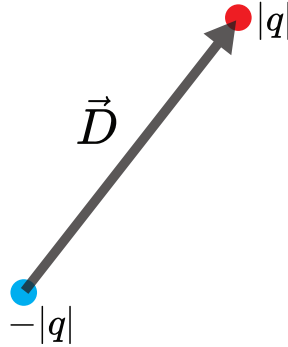


Figure 2.1: Two charges $\pm|q|$ separated by a distance $\vec{D}$.

the charge and the separation between the charges

$$\vec{d} = |q|\vec{D} \quad (2.1)$$

where q is the charge of the separated charges and $\vec{D}$ is the distance vector between them.

The relativistic interaction Hamiltonian of a neutron with a non-zero EDM reads, d_n , and a magnetic dipole moment moment, $\mu_n \approx -1.913034\mu_N$, is given by [23] ¹

$$H = H_d + H_\mu = \frac{\mu_n}{2} \bar{\psi} i \sigma_{\mu\nu} \psi F_{\mu\nu} + \frac{d_n}{2} \bar{\psi} \gamma_5 \sigma_{\mu\nu} \psi F_{\mu\nu} \quad (2.2)$$

where $F_{\mu\nu}$ is the electromagnetic field tensor. In the nonrelativistic limit this Hamiltonian can be written as

$$\begin{aligned} H &= -\mu_n \frac{\vec{s}}{s} \cdot \vec{B} - d_n \frac{\vec{s}}{s} \cdot \vec{E} \quad , \text{ or} \\ &= -\vec{\mu}_n \cdot \vec{B} - \vec{d}_n \cdot \vec{E} \end{aligned} \quad (2.3)$$

where $\vec{d}_n = d_n \vec{s}/s$ and $\vec{\mu}_n = \mu_n \vec{s}/s$. However, this interaction Hamiltonian is not

Table 2.1: Transformations of some physical quantities under **P** and **T**. + and - correspond to even and odd respectively.

Quantity	P	T
Magnetic field ($\vec{B}$)	+	-
Electric field ($\vec{E}$)	-	+
Spin ($\vec{s}$)	+	-
Magnetic dipole moment ($\vec{\mu}$)	+	-
Electric dipole moment ($\vec{d}$)	+	-

invariant under **P**, **T** and **CP**. We can see that by looking at transformation of each term in Eq. 2.3. According to table 2.1, the terms in Eq. 2.3 transform as (see also

$$^1\gamma_i = \begin{pmatrix} 0 & \sigma_i \\ -\sigma_i & 0 \end{pmatrix}, \gamma_0 = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix}, \gamma_5 = -i\gamma_0\gamma_1\gamma_2\gamma_3 \text{ and } \sigma_{\mu\nu} = \frac{1}{2} [\gamma_\mu, \gamma_\nu]$$

Fig. 2.2)

$$\begin{aligned}
\text{Under } \mathbf{P} : \vec{\mu}_n \cdot \vec{B} &\rightarrow \vec{\mu}_n \cdot \vec{B} \\
\vec{d}_n \cdot \vec{E} &\rightarrow -\vec{d}_n \cdot \vec{E} \\
\text{Under } \mathbf{T} : \vec{\mu}_n \cdot \vec{B} &\rightarrow \vec{\mu}_n \cdot \vec{B} \\
\vec{d}_n \cdot \vec{E} &\rightarrow -\vec{d}_n \cdot \vec{E}
\end{aligned} \tag{2.4}$$

Under these transformations, the interaction Hamiltonian (Eq. 2.3) is not invariant. This indicates that a permanent d_n would violate P , T symmetries. Assuming **CPT** is always conserved, **T** violation implies **CP** violation.

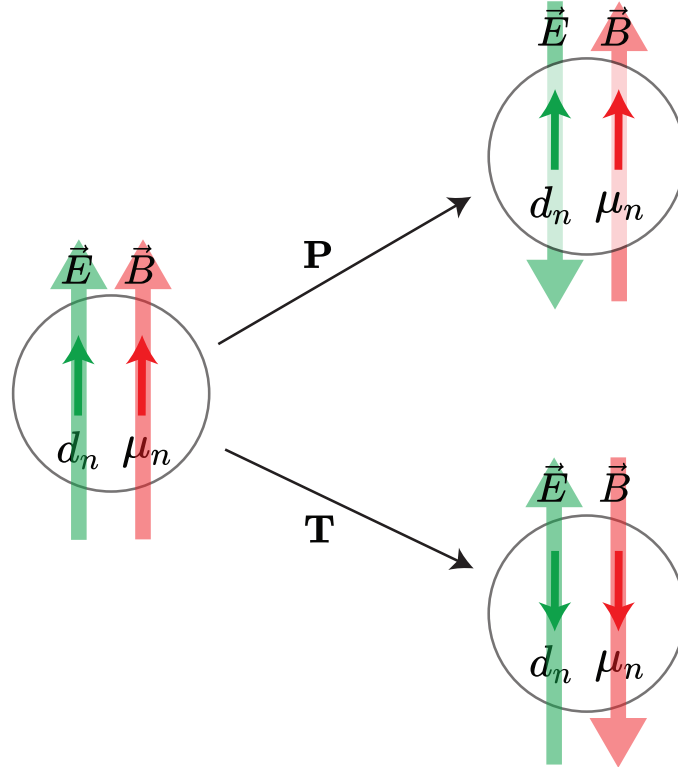


Figure 2.2: Transformation of $\vec{E}$, $\vec{B}$, $\vec{d}_n$ and $\vec{\mu}_n$ under **P** and **T** transformations.

2.1.1 Search for nEDM: New Physics Beyond Standard Model

Discovering a non-zero neutron electric dipole moment would be a significant step forward in fundamental physics. Even though neutrons are electrically neutral par-

ticles, they are nonetheless capable of possessing electric dipole moment if there is any displacement between its quark content (up and down quarks) and its center of mass. In the Standard Model, the largest contribution to the nEDM comes from the strong penguin diagram shown in Fig. 2.3. The order of this contribution is [22]

$$d_n \approx 10^{-32} e \cdot \text{cm} \quad (2.5)$$

However, this value is 6 orders of magnitude smaller than the latest global experimental limit of $d_n < 1.8 \times 10^{-26} e \cdot \text{cm}$ (90% CL) [2]. It means that any future experimental value of d_n greater than the Standard Model limits (Eq. 2.5) would indicate a new physics beyond the Standard Model. For example, some supersymmetric models are able to produce larger nEDM values than the Standard Model. A supersymmetric lower bound on the nEDM model is [4]

$$d_n \gtrsim 10^{-28} e \cdot \text{cm} \quad (2.6)$$

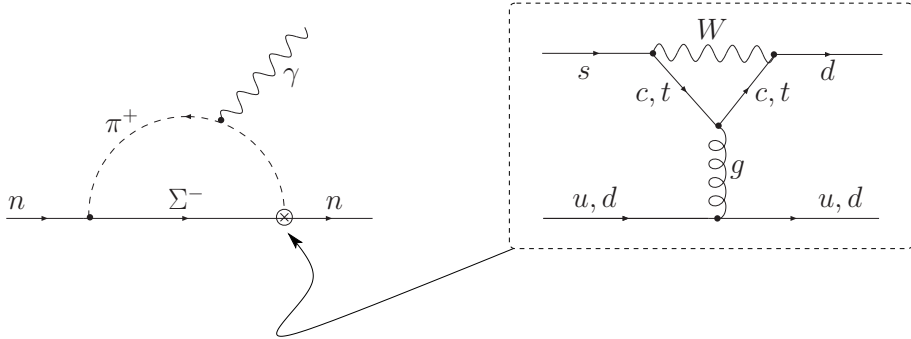


Figure 2.3: Diagram of the largest contribution to d_n in the Standard Model (Fig. from [36]).

2.1.2 Measuring the nEDM

In order to measure a non-zero neutron electric dipole moment signal, the neutrons are placed in the presence of parallel and anti-parallel electric and magnetic field configurations. The interaction given above causes the spin precess around the applied

external fields. This precession is called Larmor precession and the precession rate can be found by Eq. 2.3

$$\begin{aligned}\omega_{\uparrow\uparrow} &= \frac{2}{\hbar}(\mu_n|\vec{B}| + d_n|\vec{E}|) \\ \omega_{\uparrow\downarrow} &= \frac{2}{\hbar}(\mu_n|\vec{B}| - d_n|\vec{E}|)\end{aligned}\tag{2.7}$$

where $\uparrow\uparrow$ means $\vec{E}$ and $\vec{B}$ are parallel, and $\uparrow\downarrow$ means $\vec{E}$ and $\vec{B}$ are anti-parallel. Then, the nEDM can be extracted from the difference of the Larmor frequencies

$$d_n = \frac{\hbar\delta\omega}{4|\vec{E}|}\tag{2.8}$$

where $\delta\omega = \omega_{\uparrow\uparrow} - \omega_{\uparrow\downarrow}$.

The sensitivity limit of a nEDM measurement leveraging the Larmor precession frequency method comes from the Heisenberg uncertainty principle [23]. After a measurement period T , the uncertainty in the frequency is

$$\Delta\nu = \frac{1}{2\pi T}\tag{2.9}$$

From Eq. 2.7, we know that the frequency contribution of a non-zero nEDM is

$$\nu_d = \frac{2d_n|\vec{E}|}{\hbar}\tag{2.10}$$

For a single neutron measurement, this gives

$$\sigma_{d_n} = \frac{\hbar}{|\vec{E}|T}.\tag{2.11}$$

We can improve this uncertainty by having a large number of repeated measurements and using high number of neutrons. The statistical uncertainty for m uncorrelated different measurements and N neutrons is

$$\sigma_{d_n} = \frac{\hbar}{2e|\vec{E}|T\sqrt{mN}}.\tag{2.12}$$

According to the analysis above, we see that a large electric field and a large number of neutrons are necessary to reduce the statistical uncertainty.

2.1.3 Previous nEDM Measurements

Smith, Purcell, and Ramsey conducted the first nEDM measurement in 1957 using a neutron-beam magnetic resonance method [43]. This experiment set a limit of $d_n < 5 \times 10^{-20} e \cdot \text{cm}$. The neutron beam they utilized at Oak Ridge National Laboratory (ORNL) had a speed of $v_n \approx 2000 \text{ m/s}$. The magnetic field in the reference frame of the neutron beam is given by the Lorentz transformation

$$\vec{B}' = \gamma(\vec{B} + \frac{1}{c^2} \vec{E} \times \vec{v}) - \frac{(\gamma - 1)}{v^2} (\vec{B} \cdot \vec{v}) \vec{v} \quad (2.13)$$

where $\vec{v}$ is the neutron beam velocity and $\gamma = (1 - v^2/c^2)^{-1/2}$. For speeds much slower than the speed of light, this can be approximated as

$$\vec{B}' \approx \vec{B} + \frac{1}{c^2} \vec{E} \times \vec{v} \quad (2.14)$$

The additional field is also called the motional magnetic field:

$$\vec{B}_m = \frac{1}{c^2} \vec{E} \times \vec{v} \quad (2.15)$$

This additional field contributes to a systematic effect, the so-called geometric-phase which is proportional to the speed of the neutrons, the electric field and the magnetic field gradient.

To minimize the geometric-phase effect, ultracold neutrons (UCN) were utilized in the next generation nEDM experiments. The first nEDM experiment utilizing UCNs was conducted at the Leningrad Nuclear Physics Institute (LNPI). They achieved a nEDM limit of $d_n < 1.6 \times 10^{-24} e \cdot \text{cm}$.

The current best upper limit for nEDM has been established by the Paul Scherrer Institute (PSI). Their result was $d_n < 1.8 \times 10^{-26} e \cdot \text{cm}$. nEDM upper limits set by the previous experiments are summarized in Fig. 2.4.

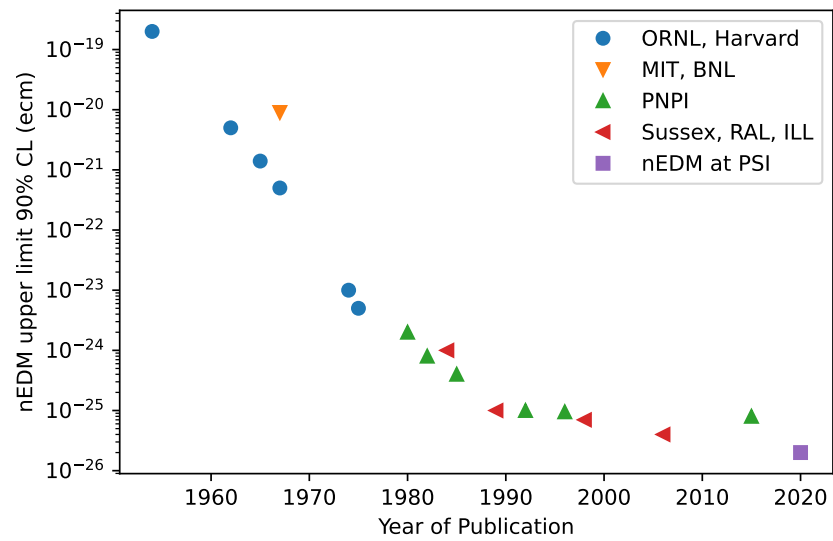


Figure 2.4: nEDM upper limits set by the previous experiments.

Chapter 3 The SNS nEDM Experiment

In this chapter, we will present an overview of the procedures that are planned to be used in the SNS nEDM experiment. The SNS nEDM experiment aims to achieve a level of sensitivity that has never been achieved before in its pursuit of measuring the neutron electric dipole moment. We will provide a high-level description of the planned experimental methods. These procedures will include novel data collecting modes that have been built expressly for such measurements. In addition, we detail how we anticipate these enhancements will result in considerable reductions in the number of systematic effects in comparison to earlier attempts made to measure the neutron dipole moment.

3.1 Measurement Methods of the SNS nEDM Experiment

The nEDM experiment, to be operated at the Spallation Neutron Source (SNS) at ORNL, is aiming to improve the current world limit by approximately two orders of magnitude by utilizing a high density of UCNs and high electric fields ($E = 75$ kV/cm). It is anticipated that this will be accomplished by downscattering the incoming neutrons in the superfluid ^4He and entrapping them inside of the measuring cells. The magnetic field inside the measurement cells will be co-monitored using polarized ^3He .

3.1.1 Production of Ultracold Neutrons

Neutrons with extremely low kinetic energies ($E \lesssim 300$ neV, $v \lesssim 8$ m/s) that can be stored inside containers are called ultracold neutrons (UCN) [38]. While higher energy neutrons can penetrate through the materials, UCNs can reflect from certain material surfaces. It is possible to create them by slowing down a stream of thermal

neutrons and scattering them off of a material having a small neutron absorption cross-section.

Let us analyze UCN - material interactions quantum mechanically. We refer to [38] for the following analysis. The scattering wavefunction for a neutron interacting with a single nucleus is given by

$$\psi(\vec{r}) = e^{ikx} + f(\theta) \frac{e^{ikr}}{r} \quad (3.1)$$

where $f(\theta)$ is the scattering amplitude and the wave vector $\vec{k}$ is along the x axis, $\vec{k} = k\hat{x}$. The de Broglie wavelength of UCNs are much larger than the interaction range of the nuclear forces R . Therefore, the scattering amplitude should be independent of θ . The resulting scattering amplitude is then

$$f(\theta) = -a \quad (3.2)$$

where a is a constant. Eq. 3.1 can be generalized to the many-nuclei case:

$$\psi(\vec{r}) = e^{ikx} - b \sum_n \psi(\vec{r}'_n) \frac{e^{ik|\vec{r}-\vec{r}'_n|}}{|\vec{r}-\vec{r}'_n|} \quad (3.3)$$

where $\vec{r}'_n$ is the position of the n -th nuclei. The right hand side of Eq. 3.3 contains a term of form $e^{ik|\vec{r}|/|\vec{r}|}$ which is the Green's function of the inhomogeneous Helmholtz equation. The Green's function $G(\vec{r}) = e^{ik|\vec{r}|}/(4\pi|\vec{r}|)$ satisfies

$$(\nabla^2 + k^2)G = -\delta(\vec{r}) \quad (3.4)$$

Thus, applying $(\nabla^2 + k^2)$ on both sides of Eq. 3.3 gives

$$(\nabla^2 + k^2)\psi = 4\pi a \left(\sum_n \delta(\vec{r} - \vec{r}'_n) \right) \psi \quad (3.5)$$

If we assume that $\lambda_n \gg |\vec{r}'_n - \vec{r}'_m|$ for any n and m , we get

$$\sum_n \delta(\vec{r} - \vec{r}'_n) \approx N \quad (3.6)$$

where N is the nuclei number density. Together with this approximation, we obtain an effective Schrodinger equation from Eq. 3.3

$$-\frac{\hbar^2}{2m}\nabla^2\psi + U_F\psi = E\psi \quad (3.7)$$

where $E = \hbar^2 k^2 / 2m$ and U_F is the Fermi potential given by

$$U_F = \frac{2\pi\hbar^2 Na}{m} \quad (3.8)$$

In order to trap UCNs inside a container, UCNs must be reflected from the surface of the material which happens when the following condition is satisfied:

$$E_{\perp} < U_F \quad (3.9)$$

where $E_{\perp} = E \sin^2 \theta$ and θ is the angle of incidence.

In the SNS nEDM experiment, incoming cold neutrons will be further slowed down to produce UCNs inside the measurement cells via neutron-superfluid ^4He interactions. This method of UCN production is called a superthermal process [16]. A neutron with energy $\hbar\omega$ and momentum $\hbar q$ can be stopped by a single downscattering if

$$\omega(q) = \frac{\hbar q^2}{2m} \quad (3.10)$$

Superfluid ^4He dispersion relation and Eq. 3.10 are illustrated in Fig. 3.1. Two curves intercept at a de Broglie wavelength of 0.89 nm or energy of $E \approx 12$ K. At this wavelength value, neutrons can come to rest and become UCNs by emitting a phonon in ^4He medium. This process is illustrated in Fig. 3.2. The probability of upscattering neutrons by phonons is verly low as the temperature of the measurement cell is around 450 mK.

3.1.2 Helium-3

The measurement cell will contain a dilute solution of polarized ^3He in superfluid ^4He . ^3He is one of the most used neutron detectors in nuclear experiments. Neutron

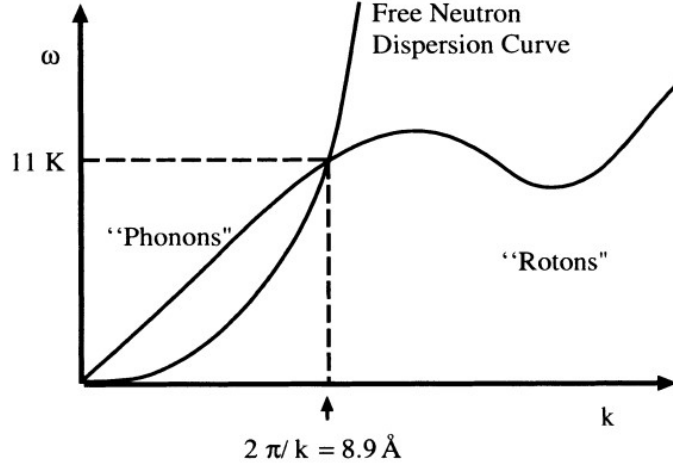


Figure 3.1: Dispersion curves for free neutrons and elementary excitations in superfluid ^4He [23].

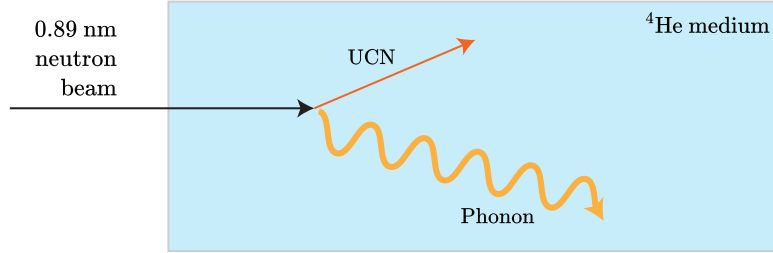


Figure 3.2: Illustration of downscattering of a neutron beam with a de Broglie wavelength of 0.89 nm in superfluid ^4He bath.

capture of ^3He is



The superfluid ^4He absorbs the energy generated by the neutron capture and it emits scintillation light with wavelength of 80 nm. This capture rate is dependent on the angle between polarization of neutrons and ^3He (θ_{n3}). The capture rate is denoted by $1/\tau_3(t)$ and is given by [6]

$$\frac{1}{\tau_3(t)} = \frac{1}{\bar{\tau}_3} (1 - P_n P_3 \cos \theta_{n3}(t)) \quad (3.12)$$

Polarizations of the UCN and ^3He are denoted by P_n and P_3 respectively. $\bar{\tau}_3$ is the average unpolarized UCN capture time; its value is ≈ 400 s for concentration $n(^3\text{He})/n(^4\text{He}) = 10^{-10}$. The scintillation light emitted is converted into visible light

by a special coating on the measurement cell walls. Then, this visible light can be detected by photomultiplier detectors.

^3He also precess under applied magnetic field as well as UCNs. They have different precession frequencies as they have different gyromagnetic ratios

$$\begin{aligned}\gamma_3 &= 2.03789 \times 10^8 \text{ rad} \cdot \text{s}^{-1} \cdot \text{T}^{-1} \\ \gamma_n &= 1.83247 \times 10^8 \text{ rad} \cdot \text{s}^{-1} \cdot \text{T}^{-1}\end{aligned}\tag{3.13}$$

where γ_3 , γ_n are the gyromagnetic ratios of ^3He and neutrons respectively. Then, we can extract the difference between UCN and ^3He precession frequencies from the scintillation signal. UCN precession rates under uniform electric and magnetic fields are given in Eq. 2.7. Similarly, the ^3He precession rate is

$$\omega_3 = \frac{2}{\hbar} \mu_3 |\vec{B}|\tag{3.14}$$

d_n can be extracted from Eq. 2.7 and 3.14,

$$d_n = \frac{\hbar [(\omega_3 - \omega_n)_{\uparrow\uparrow} - (\omega_3 - \omega_n)_{\uparrow\downarrow}]}{4|\vec{E}|}\tag{3.15}$$

where ω_n , ω_3 are precession frequencies of UCNs and ^3He respectively. Note that the individual precession frequencies are not measurable. A non-zero nEDM can be measured by measuring the difference $(\omega_3 - \omega_n)$.

In addition to the neutron capturing property of ^3He , it can also be used to measure the magnitude of the applied magnetic field inside of the measurement cell. As ^3He precess under the applied magnetic field, they will generate an oscillating magnetic field which can be detected by an array of superconducting quantum interference device (SQUID) magnetometers placed around the measurement cell. The frequency of the oscillations depends on the magnitude of the magnetic field (see Eq. 3.14). Therefore, a change in the SQUID signal would indicate a changing magnetic field.

3.1.3 Critical Dressing Method

In addition to the free precession measurement method described in previous sections, the SNS nEDM experiment anticipates using an additional measurement technique called the critical dressing method. This method involves an additional oscillating magnetic field perpendicular to the uniform magnetic field $\vec{B}_0$. This oscillating field can be used to make the gyromagnetic ratios of ^3He and neutron “equal” artificially. This would improve the background field sensitivity of the free precession method by an order of magnitude [27].

A gyromagnetic ratio γ can be “dressed” (artificially modified) by an oscillating magnetic field $\vec{B}_1$ with frequency of ω_1

$$\gamma' = \gamma J_0 \left(\frac{\gamma |\vec{B}_1|}{\omega_1} \right) \equiv \gamma J_0(x) \quad (3.16)$$

where we assumed the free precession frequency $\gamma |\vec{B}_0|$ is much smaller than ω_1 , J_0 corresponds to the Bessel function of the first kind $J_{n=0}(z)$ and $x = \gamma |\vec{B}_1| / \omega_1$ is the dressing parameter. Under an oscillating field $\vec{B}_1$, the dressed gyromagnetic ratios for ^3He and neutron are

$$\begin{aligned} \gamma'_n &= \gamma_n J_0(x_n) \\ \gamma'_3 &= \gamma_3 J_0(x_3) \end{aligned} \quad (3.17)$$

and x_n and x_3 are related by

$$x_3 = \gamma_3 \frac{x_n}{\gamma_n} \approx 1.1121 x_n \quad (3.18)$$

Then, the requirement $\gamma'_n = \gamma'_3$ implies

$$J_0(x_n) = (1.1121) J_0((1.1121)x_n) \quad (3.19)$$

which has a solution at $x_n \approx 1.189 \equiv x_c$. At this value of x_n , the ^3He and neutrons will have the same magnetic precession contribution coming from the interaction

between the magnetic field and magnetic moments. This condition is also called critical dressing.

In the case of a non-zero nEDM, the critical dressing parameter will no longer be $x_c \approx 1.189$ and the applied electric field will cause a shift in the neutron precession rate by

$$\delta\omega_n = \frac{2}{\hbar} d_n |\vec{E}| J_0(x_n) \quad (3.20)$$

As we know from Eq. 3.12, the neutron capture rate of ^3He depends on the angle θ_{n3} . Therefore, the scintillation rate will be time-dependent and this would indicate a non-zero nEDM.

3.2 SNS nEDM Measurement Cycle

In the previous section, we have described the fundamentals of the nEDM measurement methods of the SNS nEDM experiment. In this section, we will summarize a measurement cycle of the experiment step by step.

- ^3He generated in a atomic beam source is polarized via quadrupole magnet system and stored in the injection volume.
- Polarized ^3He is transported in the measurement cells. The polarization of ^3He are parallel to the applied magnetic field vector $\vec{B}_0$
- A beam of 0.89 nm cold polarized neutrons are transported to the measurement cells filled with superfluid ^4He . UCNs are produced inside the measurement cells via the downscattering process. The polarization of UCNs are parallel to the applied magnetic field vector $\vec{B}_0$ as well.
- UCN and ^3He spins are aligned in the plane orthogonal to the applied magnetic field direction (yz -plane) by a magnetic field pulse generated by a $\pi/2$ coil.

- Free precession method: UCNs and ^3He precess freely in the yz -plane. During the free precession, scintillation light produced as a result of neutron capture collected is detected by photomultipliers. The oscillating magnetic field generated by precessing ^3He isotopes is also detected by SQUID magnetometers to keep track of the magnitude of the magnetic field inside the measurement cells.
- Spin dressing method: This is an alternative of the free precession method described above. A spin dressing coil generates an oscillating magnetic field perpendicular to $\vec{B}_0$ to make the UCNs and ^3He precess with the same Larmor frequency. A time dependent scintillation rate is an indicator of a non-zero nEDM.
- After the measurement period, depolarized ^3He isotopes will be removed from the measurement cells.

3.3 SNS nEDM Apparatus Summary

3.3.1 Central Detection System

This system is made up of some of the most important measuring components, including the array of SQUID magnetometers, the high-voltage system, and the measurement cells (see Fig. 3.3). The Central Detector System (CDS) vessel is filled with liquid ^4He and all components are submerged in a liquid ^4He bath.

3.3.2 Magnetic Field System

This module includes the components required for creating a highly uniform magnetic field inside the measurement cell volume (see Fig. 3.4). A cylindrical $\cos\theta$ coil generates a uniform magnetic field ($|\vec{B}_0| = 3\mu\text{T}$) in the direction perpendicular to the cylinder axis and the neutron beam direction. A modified $\cos\theta$ coil generates an AC field to dress and rotate the ^3He and UCN spins by $\pi/2$. A set of five gradient

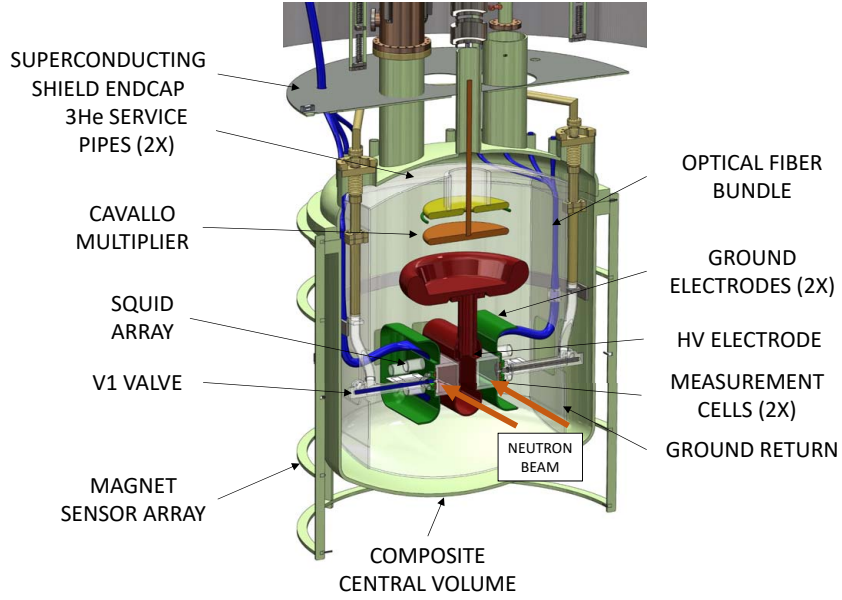


Figure 3.3: Half-section view of the CDS [6].

coils are placed onto the dressing coil to minimize any field present gradients in the magnetic field. A ferromagnetic shield (Metglas) is placed around the B_0 coil to improve field uniformity. Two sheets of copper clad G10/FR4 sheets are placed between the B_0 and the dressing coil to prevent heating due to eddy currents in the Metglas induced by the AC fields. The magnets and the shields above are surrounded by a superconducting lead layer to shield against magnetic fields changing over time.

3.3.3 Helium-3 System

This system is responsible for ^3He production and transportation. Polarized ^3He are transported from an atomic beam source (ABS) to a storage space. Then, they are moved to the measurement cells via heat flush technique. In contrast to diffusion, which is the movement of particles from a region of higher concentration to a region of lower concentration, the heat flush technique leverages the phonons in superfluid ^4He . Superfluid ^4He phonons are generated by creating a temperature gradient and they flow from hotter to colder zones. During this flow, phonons scatter from ^3He and

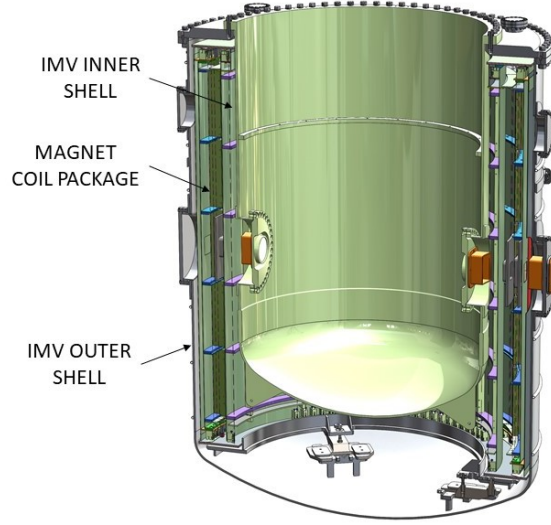


Figure 3.4: The magnet package of the SNS nEDM experiment [6].

^3He are transferred along the direction of the flow without losing polarization. Again using the heat flush technique, depolarized ^3He are removed from the measurement cells after a measurement cycle.

Chapter 4 SNS nEDM Magnetic Field Monitoring System

The cryogenic magnetic field monitoring system for the SNS nEDM experiment is introduced in [7, 31, 33, 32]. In this chapter, we present the magnetic field monitoring system with a detailed description of its components, including electronics and software. In addition, we provide initial experimental verifications of the magnetic field monitoring system performance in different magnetic field environments.

4.1 Systematic Shifts induced by Non-Uniform Magnetic Field

Even while a perfectly uniform magnetic field is theoretically possible, it is practically impossible to achieve in the measurement cells. It is possible that the precession frequency will vary due to the existence of these gradients and motional magnetic fields, resulting in a fictitious neutron electric dipole moment. In order to improve the precision of the measurement of the neutron electric dipole moment, the experiment is designed to eliminate as many causes of systematic error as feasible.

4.1.1 Spin Depolarization

Spin relaxation occurs when magnetic fields have varying magnitudes or orientations, or when they fluctuate depending on the particle's spin state. If the magnetic field inside the measurement cells is not uniform, particles that are held in place and moving within a measuring cell will be exposed to magnetic fields of varying strengths if the field within the cell is not uniform. Spins that were once aligned to each other may now be in various orientations and no longer be aligned to each other when this happens. This may be quantified as a reduction in the polarization (depolarization) of the spins. The average amount of time that it takes for spins to return to their initial orientation is referred to as the spin relaxation time. Another way to think of

this is as the average amount of time that it takes for spins that were initially aligned with each other to become randomly oriented (depolarized).

For particles in the presence of an applied magnetic field $\vec{B}_0 = B_0 \hat{x}$ and stored in a cylinder (along x -axis) of radius a and length L , the relaxation spin relaxation rate, also known as the T_2 transverse relaxation time, due to magnetic field gradients is given in [30]

$$\frac{1}{T_2} = \frac{1}{2T_1} + \frac{\gamma^2 L^4}{120D} \left(\frac{\partial B_0}{\partial z} \right)^2 + \frac{7\gamma^2 a^4}{96D} \left(\frac{\partial B_0}{\partial y} \right)^2 \quad (4.1)$$

where T_1 is the longitudinal relaxation time and D is a temperature dependent diffusion coefficient. T_1 is also given by [30]

$$\frac{1}{T_1} = \frac{\gamma^2}{2} \left[\left(\frac{\partial B_0}{\partial z} \right)^2 + \left(\frac{\partial B_0}{\partial y} \right)^2 \right] S_z(\omega_0) \quad (4.2)$$

with

$$S_z(\omega) = \frac{2k_B T \tau_c}{M\omega^2(1 + \omega^2 \tau_c^2)} \quad (4.3)$$

where T and M are the temperature and mass of the ensemble (UCN and ^3He).

4.1.2 Bloch-Siegert Induced False EDM Effect

In the case of a particle travelling in the presence of an electric field E and a holding field B_0 with non-uniformities, the particle will experience a time dependent magnetic field due to the non-uniformities and an additional magnetic field contributed by the Lorentz transformation of E . These fields will cause a shift in the precession frequency of the particle spin. This shift is called the geometric phase effect.

This effect can be illustrated by a simple example of a particle trapped in a cylindrical container along the z -axis [35]. Suppose that the particle is in the presence of a uniform magnetic gradient in addition to the uniform holding field $\vec{B}_0 = B_0 \hat{z}$ and the electric field $\vec{E} = E \hat{z}$. The magnetic field in the particle's reference frame is

$$\vec{B} = \vec{B}_r + \vec{B}_0 \quad (4.4)$$

where $\vec{B}_r$ is a rotating magnetic field with angular velocity $\vec{\omega}_r = \omega_r \hat{z}$ due to the non-uniformities in the holding field. Now, consider another reference frame moving with the particle but also rotating with the angular velocity opposite of $\vec{\omega}$. In this new reference frame, the magnetic field reads

$$\vec{B}' = B_r \hat{x}_r + \left(B_{0z} - \frac{\omega_r}{\gamma} \right) \hat{z} \quad (4.5)$$

where $\hat{x}_r$ is the x -axis of the rotating reference frame. Thus, the spin precession frequency in the rotating frame becomes

$$\omega' = \gamma B' = \gamma \sqrt{B_r^2 + \left(B_{0z} - \frac{\omega_r}{\gamma} \right)^2} \quad (4.6)$$

Deviation from the expected precession ($\omega_0 = -\gamma B_0$) frequency is then

$$\begin{aligned} \delta\omega &= \omega' - \omega_0 \\ &= \sqrt{\gamma^2 B_r^2 + (\omega_0 - \omega_r)^2} - (\omega_0 - \omega_r) \end{aligned} \quad (4.7)$$

In the $\omega_r \ll \omega_0$ limit, this shift simplifies

$$\delta\omega = \frac{\gamma^2 B_r^2}{2(\omega_0 - \omega_r)} \quad (4.8)$$

So far, we have derived a general formula for the Larmor frequency shift. Now, we can illustrate how such a shift emerges from the combination of the motional field and non-uniformities in the magnetic field. This is best illustrated for neutrons trapped in cylindrical containers. Consider a neutron moving in the plane perpendicular to the applied $\vec{E}$ and $\vec{B}_0$ (yz -plane). Moreover, it is following a circular-like path shown in Fig.4.1. Thus, we have

$$|\omega_r| = \frac{|v|}{R} \quad (4.9)$$

In the inertial reference frame of the neutron, the magnetic field gets a contribution from $\vec{E}$ due to Lorentz transformation of the electromagnetic fields. This extra field is also called the “motional field”. For $v \ll c$, it is given by

$$\vec{B}_v = \frac{\vec{E} \times \vec{v}}{c^2} \quad (4.10)$$

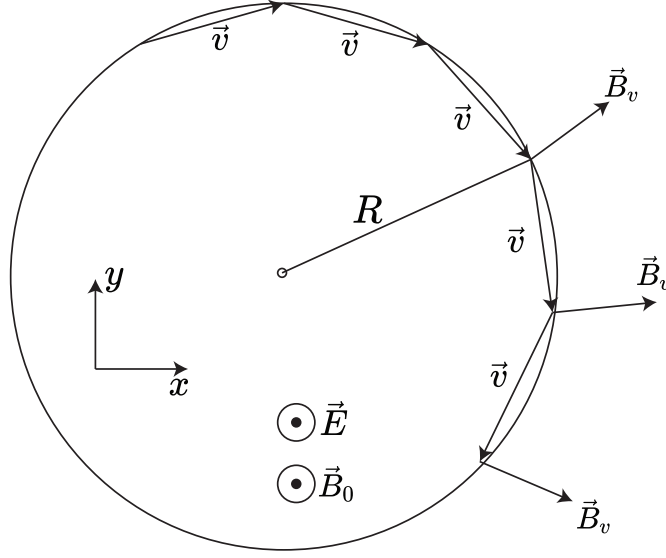


Figure 4.1: Rotation of the motional field as the particle follows a close proximity to the perimeter (Figure adapted from [35]).

Then, suppose the magnetic field non-uniformity is caused by a uniform magnetic field gradient in the vertical direction. For simplicity, we assume the magnetic field has a cylindrical symmetry. Then, from Maxwell's equations, we have

$$\frac{\partial B_{0z}}{\partial z} = -\frac{1}{2} \frac{\partial B_{0x}}{\partial x} = -\frac{1}{2} \frac{\partial B_{0y}}{\partial y} \quad (4.11)$$

Hence, the magnetic field generated by the vertical gradient is

$$\vec{B}_{0r} = -\frac{1}{2} \frac{\partial B_{0z}}{\partial z} \vec{r} \quad (4.12)$$

where $\vec{r} = (x, y)$. In the neutron frame, the net field is then

$$\vec{B} = -\frac{1}{2} \frac{\partial B_{0z}}{\partial z} \vec{r} + \frac{\vec{E} \times \vec{v}}{c^2} \quad (4.13)$$

The frequency shift for $\vec{B}_0$ and $\vec{E}$ fields parallel and anti-parallel can be obtained by substituting the total magnetic field into Eq. 4.8

$$\delta\omega_{\uparrow\uparrow, \uparrow\downarrow} = \frac{\gamma^2 \left(-\frac{1}{2} \frac{\partial B_{0z}}{\partial z} \vec{r} \pm \frac{\vec{E} \times \vec{v}}{c^2} \right)^2}{2(\omega_0 - \omega_r)} \quad (4.14)$$

In a nEDM measurement, we are interested in the difference in the frequency shift for $\vec{B}_0$ and $\vec{E}$ fields parallel and anti-parallel. This difference is

$$\delta\omega_{\uparrow\uparrow} - \delta\omega_{\uparrow\downarrow} = -\gamma^2 R \frac{\partial B_{0z}}{\partial z} \frac{vE|\omega_r|}{c^2(\omega_0^2 - \omega_r^2)} \quad (4.15)$$

Since UCNs are extremely slow particles, we can assume $\omega_r \ll \omega_0$. In this approximation we have

$$\delta\omega_{\uparrow\uparrow} - \delta\omega_{\uparrow\downarrow} \approx \frac{\partial B_{0z}}{\partial z} \frac{v^2 E}{2c^2 B_{0z}^2} \frac{\omega_0^2 \omega_r^2}{\omega_0^2 - \omega_r^2} \quad (4.16)$$

Eq. 4.16 is basically equation (19) in [35], which is

$$d_{af} = -\frac{J\hbar}{2} \left(\frac{\partial B_{0z}/\partial z}{B_{0z}^2} \right) \frac{v_{xy}^2}{c^2} \left[1 - \frac{\omega_r^2}{\omega_0^2} \right]^{-1} \quad (4.17)$$

4.2 Magnetic Field Prediction Method

It is not always practicable or practical to monitor the magnetic field in real time in a particular region of space that is designated for experimental purposes. It may be the situation that installing a magnetic field sensor inside the experimental zone would cause too much of a disruption to the system, or it may be the case that the experimental region is inaccessible owing to a physical enclosure. For instance, an experiment could require a magnetic field that is extremely uniform. The field created by the coil would most likely be measured directly using a field probe in order to verify that the coil was designed appropriately. The coil would need to be designed and constructed in order to produce such a field. However, once the apparatus has been assembled, it is conceivable that it will not be feasible to measure the field directly (for instance, if there isn't enough room to put a probe in different spots). In these cases, indirect measurements will need to be taken instead. If it is necessary to monitor the field during an experiment while it is being carried out, then this presents a hurdle. The SNS nEDM experiment is confronted with exactly this difficulty. The central region of the magnet in which the measurements are taken is inaccessible to field probes, so indirect methods must be used instead.

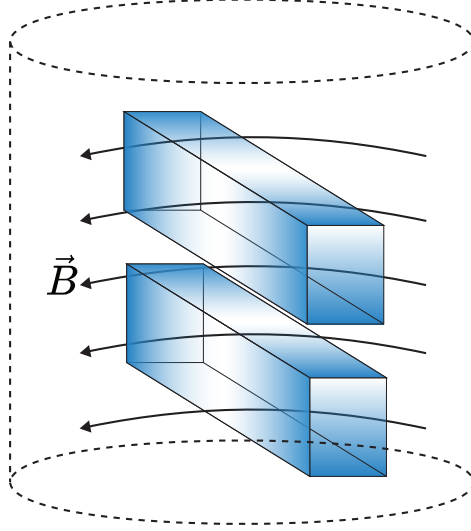


Figure 4.2: A simplified graphical representation of the concept. The magnetic field components within an inaccessible interior region of the experiment can be inferred from the measurements of the vector field components on a surface that is located outside of the experimental volume (dashed line).

Fig. 4.2 illustrates how measurements of the field components may be performed at several positions over an outside surface. If there are no free currents within the contained volume, then it follows

$$\vec{\nabla} \times \vec{B} = 0 \quad (4.18)$$

Eq. 4.18 implies that we can define a magnetic scalar potential Φ_M

$$\vec{B} = -\vec{\nabla}\Phi_M \quad (4.19)$$

Furthermore, if we also assume there is no source of magnetization or currents within the contained volume, i.e. $\vec{M} = 0$, $\vec{\nabla} \cdot \vec{B} = 0$ and Eq. 4.18 gives a Laplace equation for the magnetic scalar potential,

$$\nabla^2 \Phi_M = 0 \quad (4.20)$$

Analytical solutions of Eq. 4.20 are well known and they include harmonic $\sim 1/r^{l+1}$ and $\sim r^l$ terms. We are interested in the region which include the $r = 0$ point, and

at that point $1/r \rightarrow \infty$. Therefore, the solution concerning the interior space is given as

$$\Phi_M(r, \theta, \phi) = \sum_l \sum_{m=0}^l r^l P_l^m(\cos \theta) [a_{lm} \cos(m\phi) + b_{lm} \sin(m\phi)] \quad (4.21)$$

where a_{lm}, b_{lm} are expansion coefficients and P_l^m are the Legendre polynomials. From Eq. 4.19, the magnetic field components can be found [33]

$$\begin{aligned} B_r(\vec{r}) &= -\frac{1}{r} \sum_{l,m} [l C_{lm}(\vec{r}) a_{lm} + l S_{lm}(\vec{r}) b_{lm}] \\ B_\theta(\vec{r}) &= -\frac{1}{\sin \theta} \sum_{l,m} [\Delta_{lm}(\vec{r}) a_{lm} + \Lambda_{lm}(\vec{r}) b_{lm}] \\ B_\phi(\vec{r}) &= -\frac{1}{r \sin \theta} \sum_{l,m} [m C_{lm}(\vec{r}) a_{lm} - m S_{lm}(\vec{r}) b_{lm}] \end{aligned} \quad (4.22)$$

where

$$\begin{aligned} C_{lm}(\vec{r}) &= r^l P_l^m \cos \theta \cos(m\phi) \\ S_{lm}(\vec{r}) &= r^l P_l^m \cos \theta \sin(m\phi) \\ \Delta_{lm}(\vec{r}) &= -\frac{l}{r} C_{lm}(\vec{r}) \cos \theta - (l+m) C_{l-1,m}(\vec{r}) \\ \Lambda_{lm}(\vec{r}) &= -\frac{l}{r} S_{lm}(\vec{r}) \cos \theta - (l+m) S_{l-1,m}(\vec{r}) \end{aligned} \quad (4.23)$$

We can rewrite Eq. 4.22 in a compact form in Cartesian coordinates as

$$\vec{B}(x, y, z) = \sum_i a_i \vec{f}_i(x, y, z) \quad (4.24)$$

where $\vec{f}_i(x, y, z)$ are Cartesian basis vector functions and a_i are expansion coefficients to be determined. In order to determine the values of the coefficients a_i to a certain order $i = N$, we would need to measure a number of magnetic field components, denoted by M and each measurement is along a single direction j_k . Note that the summation on the right hand side is a series summation. Then, the expansion coef-

ficient can be obtained by solving a system of linear equations ¹

$$\begin{aligned}
B_{j_1}(\vec{r}_1) &= \sum_{i=1}^N a_i f_{j_1,i}(\vec{r}_1) \\
B_{j_2}(\vec{r}_2) &= \sum_{i=1}^N a_i f_{j_2,i}(\vec{r}_2) \\
&\vdots \\
B_{j_M}(\vec{r}_M) &= \sum_{i=1}^N a_i f_{j_M,i}(\vec{r}_M)
\end{aligned} \tag{4.25}$$

where $f_{j_k,i}$ denotes the j_k component of the basis vector $\vec{f}_i$ and $\vec{r}_1, \vec{r}_2, \dots, \vec{r}_M$ are the locations of the discrete magnetic field measurements. This system of linear equations can be written in terms of matrices

$$\begin{pmatrix} B_{j_1}(\vec{r}_1) \\ B_{j_2}(\vec{r}_2) \\ B_{j_3}(\vec{r}_3) \\ B_{j_4}(\vec{r}_4) \\ \vdots \\ B_{j_M}(\vec{r}_M) \end{pmatrix} = \begin{pmatrix} f_{j_1,1}(\vec{r}_1) & f_{j_1,2}(\vec{r}_1) & f_{j_1,3}(\vec{r}_1) & \dots & f_{j_1,N}(\vec{r}_1) \\ f_{j_2,1}(\vec{r}_2) & f_{j_2,2}(\vec{r}_2) & f_{j_2,3}(\vec{r}_2) & \dots & f_{j_2,N}(\vec{r}_2) \\ f_{j_3,1}(\vec{r}_3) & f_{j_3,2}(\vec{r}_3) & f_{j_3,3}(\vec{r}_3) & \dots & f_{j_3,N}(\vec{r}_3) \\ f_{j_4,1}(\vec{r}_4) & f_{j_4,2}(\vec{r}_4) & f_{j_4,3}(\vec{r}_4) & \dots & f_{j_4,N}(\vec{r}_4) \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ f_{j_M,1}(\vec{r}_M) & f_{j_M,2}(\vec{r}_M) & f_{j_M,3}(\vec{r}_M) & \dots & f_{j_M,N}(\vec{r}_M) \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \\ \vdots \\ a_N \end{pmatrix} \tag{4.26}$$

or, shortly

$$\mathbf{B} = \mathbf{F}\mathbf{A} \tag{4.27}$$

Note that $\mathbf{F}$ is a $(M \times N)$ matrix. If the number of basis functions, N , is equal to the number of measurements, M , Eq. 4.27 has a simple solution

$$\mathbf{A} = \mathbf{F}^{-1}\mathbf{B} \tag{4.28}$$

However, when $M \neq N$, the basis function matrix $\mathbf{F}$ does not have an inverse. In this case, we can solve this problem in the “best way possible” by introducing a generalized inverse of a matrix that applies to rectangular matrices as well. The

¹Here j_M denotes one of the Cartesian components; $j_k \in \{x, y, z\}$.

impact of data inaccuracies can be mitigated by using more observations than there are model parameters that are currently uncertain. When $M > N$, an overdetermined linear system of equations can be used to describe the issue. If $M < N$, the least squares solution does not have a unique solution. In that case, Eq. 4.28 gives a minimum norm solution.

One such inverse is the Moore-Penrose inverse (or the pseudoinverse). It can be used to find the least squares solution to a system of linear equations such as Eq. 4.27 when $M \neq N$. The pseudoinverse of matrix $\mathbf{F}$ can be constructed by the singular value decomposition (SVD). The SVD decomposition of $\mathbf{F}$ is given as

$$\mathbf{F} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^* \quad (4.29)$$

where $\mathbf{\Sigma}$ is a diagonal matrix and $\mathbf{U} \in \mathbf{R}^{M \times M}$ and $\mathbf{V} \in \mathbf{R}^{N \times N}$ are orthogonal matrices. The pseudoinverse $\mathbf{F}^+$ is defined as

$$\mathbf{F}^+ = \mathbf{V}\mathbf{\Sigma}^+\mathbf{U}^* \quad (4.30)$$

where $\mathbf{\Sigma}^+$ is the matrix constructed by taking the reciprocal of all non-zero elements of $\mathbf{\Sigma}^T$. Finally, the least squares solution of the smallest norm is given by

$$\mathbf{A}_{\text{best}} = \mathbf{F}^+\mathbf{B} \quad (4.31)$$

4.3 Magnetic Field Monitor System

4.3.1 Probe Array Structure

The probe array structure was designed to contain 39 single-axis fluxgate magnetic field probes (Bartington Mag F) (Fig. 4.3). This design allows for the probes to be evenly spaced around the circumference of the cylinder, with each probe pointing in either the x , y or z directions. The overall height of the array structure is 48 inches, with a radius of 24.5 inches. The weight of the entire structure is approximately 100 lbs. The structure is made of G10/FR-4, which is a composite made from glass cloth



Figure 4.3: Completely built magnetic field monitor probe array with Bartington single axis fluxgates.

with an epoxy resin binder. It has excellent dielectric properties and a high mechanical strength. It is typically used in applications where reliability and dimensional stability are important.

The probes are placed in custom-made Torlon holders, which are mounted on the main structure using PEEK screws and nuts. The Torlon holders provide a high level of mechanical stability, while still allowing for the probes to be easily removed and replaced. All material used in the probe array structure have a low thermal expansion coefficient, which helps to minimize measurement errors due to temperature changes (Fig. 4.4).

4.3.2 Single Axis Fluxgate Magnetometer

The cryogenic-compatible single-axis fluxgate magnetometer system we utilize is used to measure the strength and direction of a magnetic field along one axis. This type

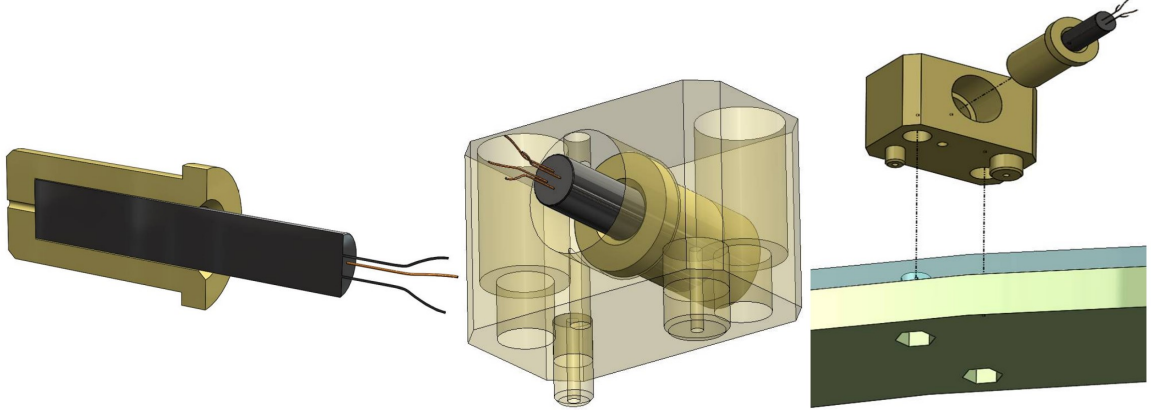


Figure 4.4: Single axis probe assembly. The probes are glued in Torlon bushings and mounted by sliding them into a hole inside the holder. The probe is then locked in place by a PEEK setscrew.

of magnetometer typically consists of two parts: a probe that measures the magnetic field (Mag F, a cryogenic-compatible single axis fluxgate probe), and a processing unit that interprets the input from the sensor (Mag-01H). The range and resolution properties of the magnetometer system is given in Table 4.1.

Table 4.1: Measurement range and resolution table of the Mag F probe.

Mag-01H Sensitivity	Range (μT)	Resolution (nT)
$\times 1$	0-20	1
	20-290	10
$\times 10$	0-2	0.1
	2-100	1
	100-290	10

4.3.3 Data Acquisition System

The data collection process is illustrated in Fig. 4.5. We have designed a probe switching system that allows the user to pick which single axis probe input they would want to utilize at a particular moment and then delivers that input signal into a single line. This is advantageous because it enables the user to pick whatever probe input they wish to utilize without physically moving or switching cables. This is accomplished through the use of EMI protected reed relays connected to each of

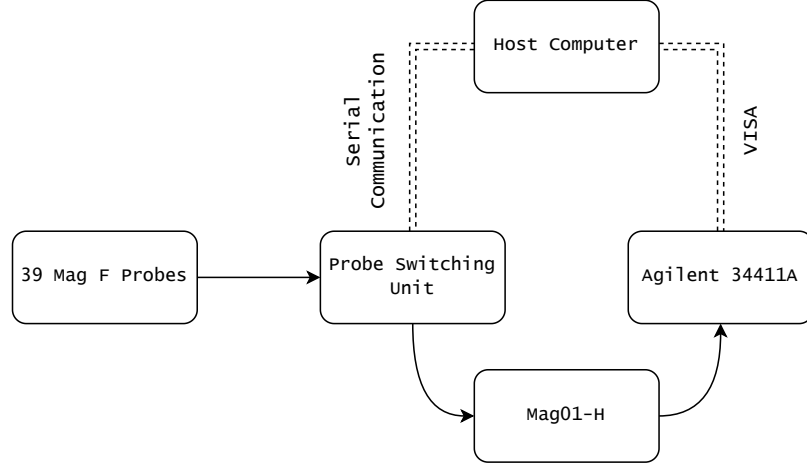


Figure 4.5: Schematics of the magnetic field system.

the probe inputs. The reed relays are activated by an Arduino-controlled 16-port constant-current sink LED driver.

This system’s user interface is a Python script that talks with Arduino using serial communication. The Arduino is configured to continuously check for serial input from the Python script and alter the state of the reed relays based on this data.

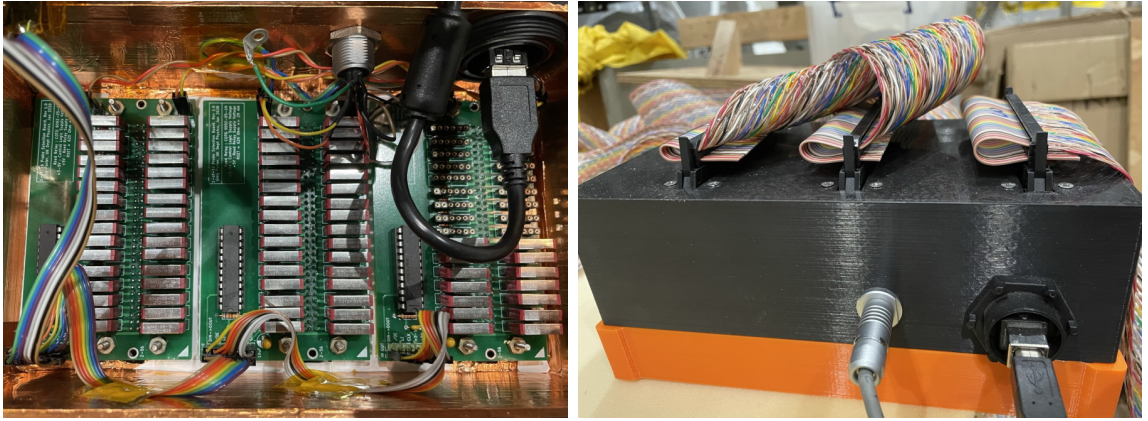


Figure 4.6: Left: Printed circuit boards and components of the probe switching system. Right: Outside view of the system.

4.4 Tests of the Magnetic Field Monitor

In this section, we demonstrate the initial test results of the magnetic field monitor system at room and cryogenic temperatures.

4.4.1 Room Temperature Tests at UKY

This test was carried out at the University of Kentucky in order to verify that the system was working properly; the magnetic field monitor was initially examined when it was at room temperature. The probe array was placed in a pair of anti-Helmholtz coils, so as to be subject to an ambient field and several different strengths of additional applied magnetic field gradients in the x direction. Two of the coils were placed along the x -axis. The experimental setup is shown in Fig. 4.7. A triple-axis



Figure 4.7: Room temperature test setup of the magnetic field monitor. A triple-axis magnetometer was installed on an automatic three-dimensional mapping system to test the performance of the magnetic field monitor.

magnetometer (Bartington Mag-03MSESL100) was installed on an automatic three-dimensional mapping system to test the performance of the magnetic field monitor. This was done so as to determine whether or not the reconstruction was accurate. The triple-axis magnetometer probe was positioned at the center of the probe array volume, and passed through a neutron beam window. The measurements were taken while moving the magnetometer along the x , y and z axes. The results are plotted in Figs. 4.8-4.14 (see Appendix B for numerical values), which shows the reconstruction results when different magnetic field gradients were applied. We were able to capture the changes in the field gradient with a magnitude of $\sim 2 \times 10^{-5}$ G/cm.

Table 4.2 summarizes the differences in average gradients along x axis for different anti-Helmholtz current values. The results show that the proposed magnetic field monitor can be used to accurately measure the gradient of a magnetic field.

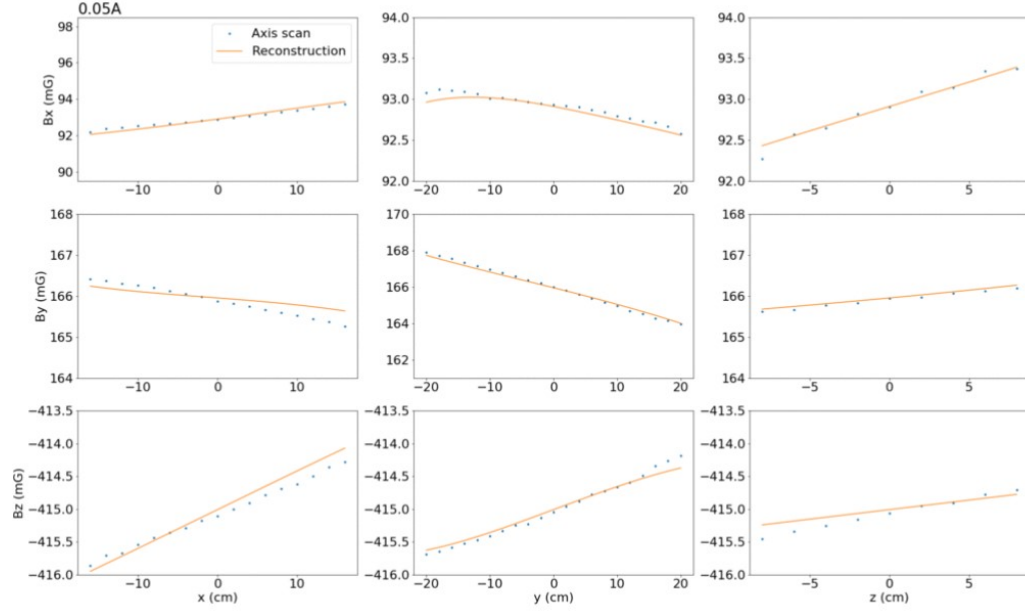


Figure 4.8: Comparison of the reconstruction results (orange line) and the magnetometer scan along x, y and z axes (blue dots). Anti-Helmholtz coils current: $I = 50\text{ mA}$.

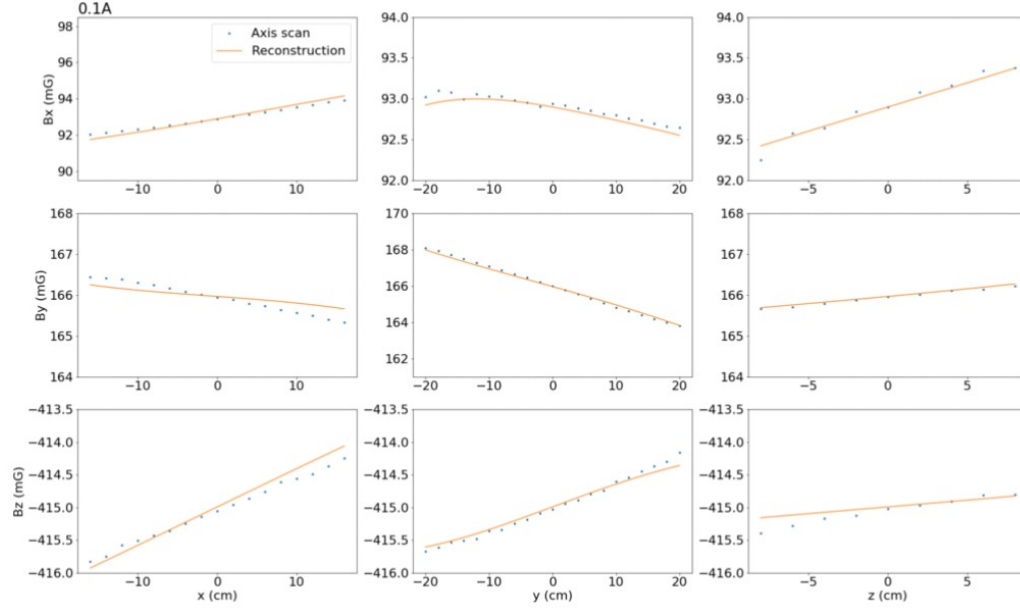


Figure 4.9: Comparison of the reconstruction results (orange line) and the magnetometer scan along x, y and z axes (blue dots). Anti-Helmholtz coils current: $I = 100$ mA.

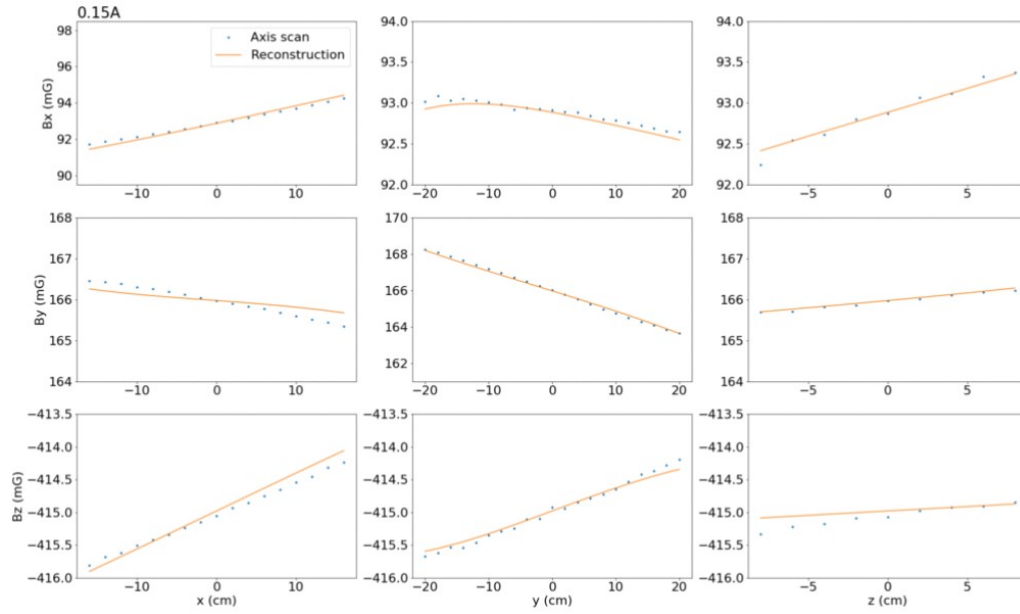


Figure 4.10: Comparison of the reconstruction results (orange line) and the magnetometer scan along x, y and z axes (blue dots). Anti-Helmholtz coils current: $I = 150$ mA.

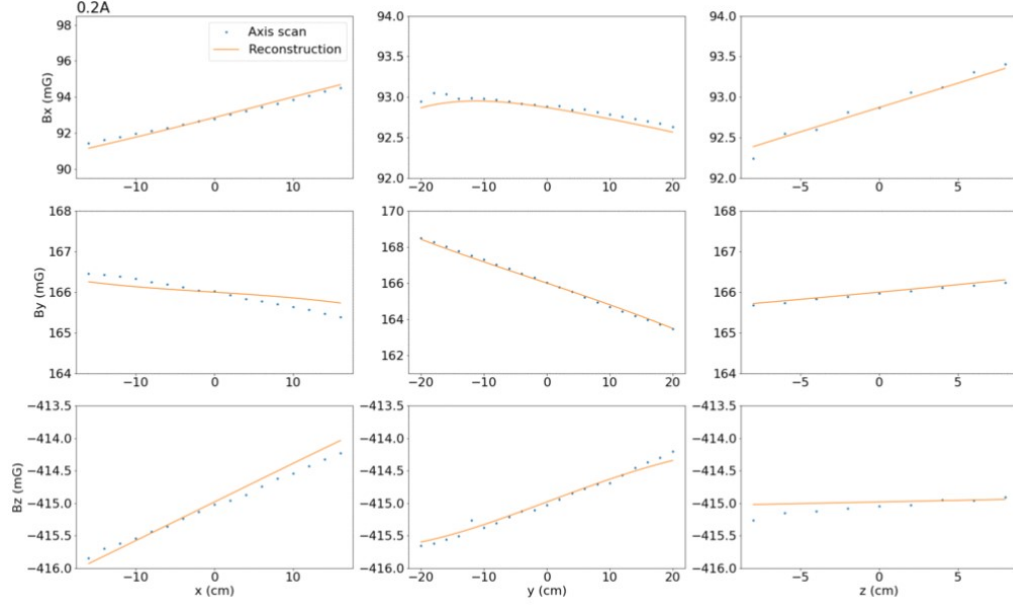


Figure 4.11: Comparison of the reconstruction results (orange line) and the magnetometer scan along x, y and z axes (blue dots). Anti-Helmholtz coils current: $I = 200$ mA.

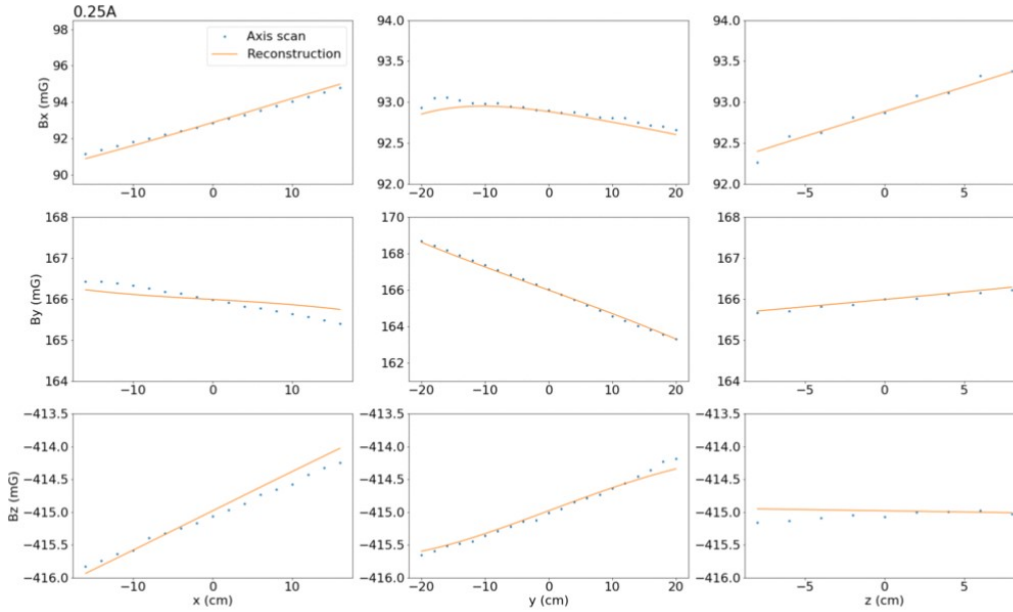


Figure 4.12: Comparison of the reconstruction results (orange line) and the magnetometer scan along x, y and z axes (blue dots). Anti-Helmholtz coils current: $I = 250$ mA.

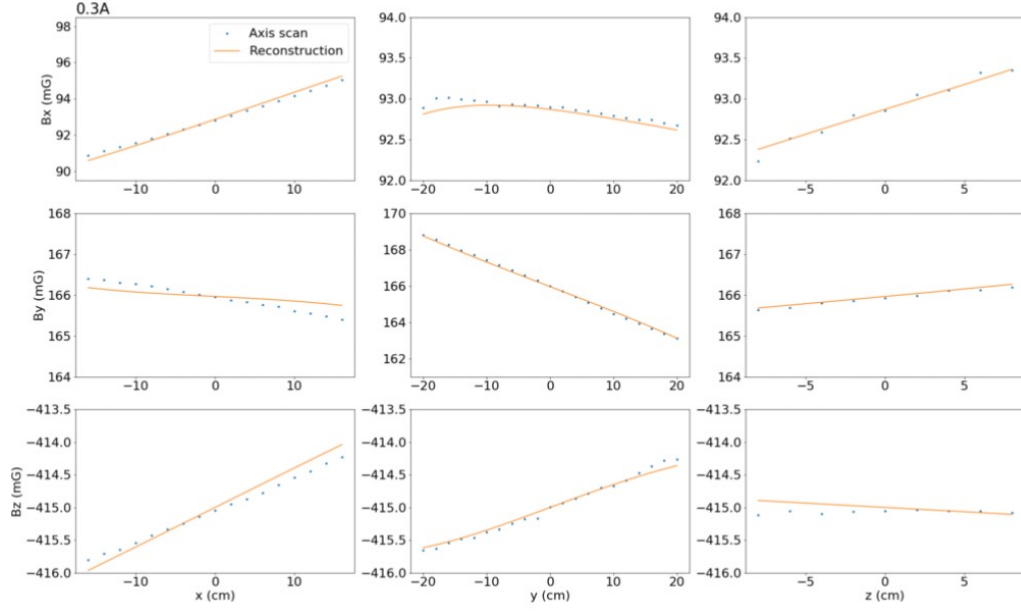


Figure 4.13: Comparison of the reconstruction results (orange line) and the magnetometer scan along x , y and z axes (blue dots). Anti-Helmholtz coils current: $I = 300\text{ mA}$.

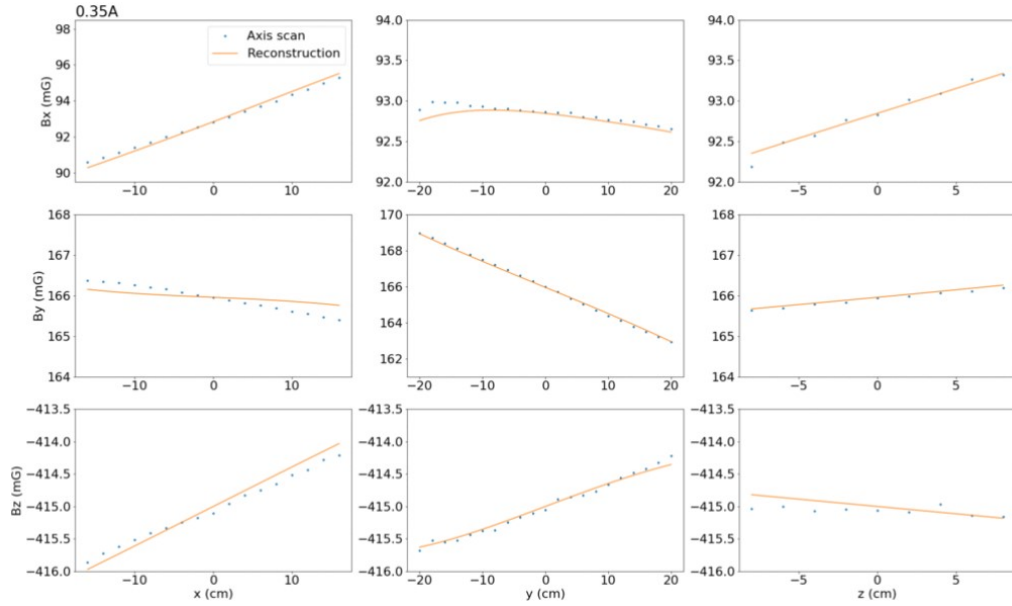


Figure 4.14: Comparison of the reconstruction results (orange line) and the magnetometer scan along x , y and z axes (blue dots). Anti-Helmholtz coils current: $I = 350\text{ mA}$.

Table 4.2: Differences in average gradients along x axis, $\langle dB_x/dx \rangle$ for different anti-Helmholtz current values.

Difference	Value [G/cm]
$\langle dB_x/dx \rangle_{0.1A} - \langle dB_x/dx \rangle_{0.05A}$	2.169×10^{-5}
$\langle dB_x/dx \rangle_{0.15A} - \langle dB_x/dx \rangle_{0.1A}$	2.099×10^{-5}
$\langle dB_x/dx \rangle_{0.2A} - \langle dB_x/dx \rangle_{0.15A}$	2.151×10^{-5}
$\langle dB_x/dx \rangle_{0.25A} - \langle dB_x/dx \rangle_{0.2A}$	2.285×10^{-5}
$\langle dB_x/dx \rangle_{0.3A} - \langle dB_x/dx \rangle_{0.25A}$	2.361×10^{-5}
$\langle dB_x/dx \rangle_{0.35A} - \langle dB_x/dx \rangle_{0.3A}$	2.525×10^{-5}

4.4.2 Magnet Test Setup at Caltech

I relocated to the California Institute of Technology to do experimental work, and I worked with Dr. Brad Filippone's group on installing and testing a prototype system in a magnet testing apparatus at Caltech. I was actively involved in both of these processes from December 2020 to May 2022.

In the course of my work at Caltech, I was principally responsible for installing the magnetic field monitor and evaluating its operation under cryogenic conditions. The B_0 coil needed to be installed inside the cryostat as the initial stage in the process. Within the cryostat was a simple platform that served as a support for the coil. After the B_0 coil had been installed, the probe array construction was suspended on a G-10 support hoop that was situated under the lid of the cryostat. The vacuum feedthrough sent the 39 probe wires from the cryostat to the air side, where they were linked to the probe switching circuit via three sets of twisted pair ribbon cables. The ribbon cable connections were secured by the strain relief provided by the flat cable type rectangular socket connector. The Caltech magnet test setup is shown in Fig. 4.15.

Before the measurement, the cryostat was cooled down and the temperature of the probe array structure was measured with a Lakeshore diode temperature sensor. The volume where the probe array was located reached a minimum temperature of ~ 2 K. A Bartington Mag-03MCTPL70 three-axis field probe is used to collect the

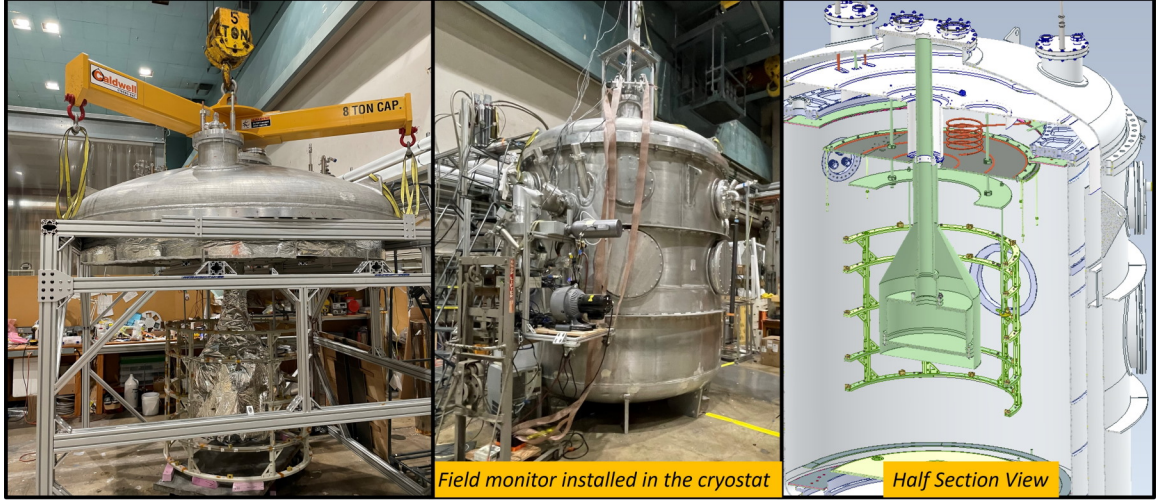


Figure 4.15: Magnet test setup at Caltech. The magnetic field monitor array is placed at the center of the magnet volume to determine the magnetic field profile of the B_0 coil.

data for the measurements. This probe is installed on a table made of aluminum, and numerous encoder motors are utilized to translate, rotate, and raise and lower the probe arm (Fig. 4.16). The region of interest for field mapping is the center of the magnet and extending outward by 10-20 cm in all directions. In its initial configuration, the mapper was installed on a table that was attached to the top of the SNS nEDM cryostat. We made use of a script that was built in LabVIEW in order to carry out automated control of the mapper arm as well as a DC power supply that drives the B_0 coil. The script was run with a fixed number of points to map, and we set the power supply to operate in constant current mode before doing so. In order to reduce background noise in measurements caused by time-varying magnetic fields, we gathered data in successive runs while the power supply was turned on, then off. This allowed us to do background subtraction. In addition, we let the mapping application run overnight, which enabled us to obtain data with a reduced amount of background noise.

As can be seen in Fig. 4.17 (see Appendix B for numerical values), we mapped the B_0 magnetic field (~ 33 mG) across a distance of 36 centimeters along the x



Figure 4.16: An image of the finished mapper arm setup, mounted on the SNS nEDM cryostat.

axis and then used the data from the magnetic field monitor to reconstruct the field. In addition to this, we used a COMSOL model provided by Christopher Swank to perform a simulation of the magnetic field that was produced by the B_0 coil and the simulation result was compared to the result of the magnetic field reconstruction. The linear gradient extracted from the COMSOL model, 6.7×10^{-6} G/cm, and the reconstructed linear gradient, 7.2×10^{-6} G/cm, were found to be in good agreement.

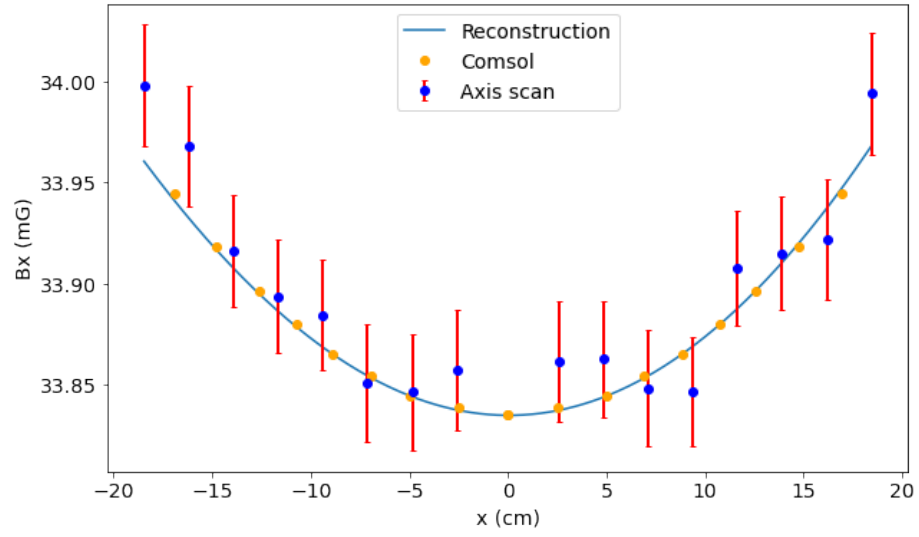


Figure 4.17: Comparison of magnetic field mapper reconstruction, COMSOL model and magnetic field scan of the field generated by the B_0 coil at liquid nitrogen temperatures.

Chapter 5 Conclusion

In summary, we constructed and tested a magnetic field monitor to measure the magnetic field at cryogenic temperatures for the SNS nEDM experiment. The design consists of 39 single-axis magnetometers located on a cylindrical spaceframe that allow for the collection of data from each 39 discrete locations and then reconstructing the magnetic field. The preliminary data collected in the earlier sections 4.4.1 and 4.4.2 demonstrate that the reconstruction findings of the magnetic field monitor are in excellent agreement with both the magnetic field scanning and the COMSOL simulation. In the room temperature test of the field monitor, we achieved to capture the changes in the field gradient with a magnitude of $\sim 2 \times 10^{-5}$ G/cm. Moreover, in the cryogenic temperature tests performed at the California Institute of Technology, the linear gradient extracted from the COMSOL model, 6.7×10^{-6} G/cm, and the reconstructed linear gradient, 7.2×10^{-6} G/cm, were found to be in good agreement. The requirement for the magnetic field monitor is to be able to detect field gradients to the level of $\sim 10^{-6}$ G/cm and we have shown that we are able to meet this goal.

Shipping the magnetic field monitor to Oak Ridge National Laboratory (ORNL) is the next stage in the process, which will be followed by monitoring the field in cryogenic conditions for the polarization transmission test and, eventually, the final configuration.

Chapter 6 Magnetic Field Mapping Using Physics-Informed Neural Networks

Developments in this chapter are based on the article [13], published in Scientific Reports, coauthored with Bilgehan Sel and Brad Plaster.

6.1 Introduction

Magnetic field mapping is commonly used in many fields of science, medicine and technology such as particle accelerators, nuclear storage experiments [6, 5, 17], cardiac beat detection [39], magnetic resonance imaging (MRI) [18] and magnetic indoor positioning systems (IPS) [28, 19]. For example, in nuclear and particle physics experiments, with one example being the search for the neutron electric dipole moment, it is often crucial to measure and control the magnetic field components in the experimental region, because these experiments are typically sensitive to perturbations in magnetic fields. An undetected disturbance in a magnetic field may result in systematic uncertainties and cause a limitation for the precision of the measured quantities. To minimize systematic uncertainties, magnetic field components should be monitored in real-time and any unwanted field should be compensated during the operation time of the experiment. Real-time measurement of the magnetic field in an experimental region of space is not always practical or feasible. In most cases, the experimental region is not accessible due to a physical enclosure (e.g., a setup placed in a vacuum chamber), or it could be that placing a magnetic field sensor inside the experimental region is too disruptive to the system.

There exist several approaches in the literature that can be utilized to solve the problems stated above. For instance, Solin et al. [44] make use of Gaussian processes (GPs) to interpolate/extrapolate ambient magnetic fields. They train the model us-

ing a data set collected by a magnetic field sensor at different locations of space and reconstruct the whole ambient magnetic field. Another method is proposed by Nouri et al. [32, 33] (see Chapter 4 for detailed description). They introduced a non-disruptive magnetic field mapping method using exterior measurements at fixed locations and leverage the multipole expansion of the magnetic field vector. Expanding the magnetic field to some finite degree $n = N$, they provide a systematic way to optimize sensor locations and fit the unknown coefficients of the multipole expansion using the data from those exterior sensor measurements. This method is susceptible to noise in the data and, since the multipole expansion terms need to be picked to a specific field profile, the coefficients of the expansion terms are not regularized.

In this chapter, we propose a robust way of predicting the magnetic field vector in the experimental region. In order to accomplish this, we utilize physics-informed neural networks (PINNs) [37]. PINNs propose a way to incorporate prior physical knowledge about the system in terms of its partial differential equations, into the deep neural networks while still being able to utilize their universal function approximator property. With PINNs, data and mathematical models of physics are combined in a smooth way, even in situations that are only partially understood, uncertain, and have a lot of dimensions. In noisy and high-dimensional situations, physics-informed learning blends data and mathematical models easily and can solve general inverse problems extremely successfully [42, 41, 15]. Unlike the method proposed in [32, 33], our method does not require prior knowledge of the multiple expansion terms to be fitted. This special type of neural network regularizes the output function (magnetic field prediction) during the training process by requiring the output to satisfy Maxwell’s equations, specifically Ampere’s law and Gauss’ law for magnetism.

6.2 Artificial Neural Networks

An artificial neural network (ANN) is a machine learning algorithm that is used to model complex patterns in data. Neural networks are comparable to other machine learning methods, but they are comprised of a huge number of interconnected nodes (neurons), which may learn to detect input data patterns. ANNs have been put to use in a wide variety of applications, including facial recognition, the recognition of handwritten characters or digits, and a great deal of other work involving the detection of patterns. The use of neural networks is a strong technique that may be utilized to find solutions to difficult issues.

A neural network is composed of three types of layers: input, hidden, and output. The input layer is where the data is fed into the network. This data can be in the form of numbers, images, or any other type of data that can be processed by a computer. When the neural network is run, its results are sent out through what's called the output layer. This can be in the form of a classification (e.g. is this an image of a dog or a cat?), a prediction (e.g. what will the stock market do tomorrow?), or anything else that the neural network has been trained to output. The hidden layers are where the actual processing of the data takes place. This is where the artificial neurons are located and where the magic happens! There is no hard and fast rule that dictates how many hidden layers should be employed in a neural network; therefore the number of these layers can change as needed. A neural network with a single hidden layer can approximate any function that you want, but adding more layers allows the algorithm better approximation of more complicated functions. In general, the network's ability to approximate functions will improve in proportion to the number of hidden layers. On the other hand, if there are an excessive number of hidden layers, the network faces the danger of overfitting the data. This indicates that it has a poor ability to generalize to new data since it has an overly strong memory for individual training data points, and not the overall trend. When building neural networks, machine

learning engineers need to be cognizant of the trade-off between underfitting and overfitting, which may be summarized as “the balance between too little and too much.”

A node, often known as a neuron, in an artificial neural network is responsible for both receiving input from other nodes and transmitting output to other nodes (Fig. 6.1). When modeling artificial neurons, the intricacy of genuine neurons is greatly simplified and abstracted. These are essentially made up of inputs, which are then multiplied by weights before being added together. Following that, the result is sent to an activation function (such as a sigmoid), and only the output of that function is used for training purposes. There are many other kinds of activation functions, but the rectified linear unit (ReLU) and the softmax activation function are the ones that are most commonly used in deep learning. The “weight” of an artificial neuron is a

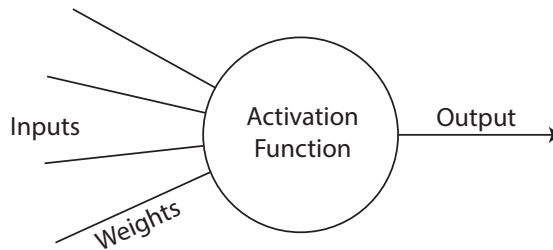


Figure 6.1: Illustration of an artificial neuron (or node).

parameter that is learned by the neural network during training. It is used to control the strength of the connections between neurons in the network. When the weight is negative, the signal is said to be suppressed. The computations that are produced by a neuron are uniquely influenced by its weights. We are able to tune a certain output for a set of inputs by only modifying the weights. However, if there are hundreds or thousands of neurons included inside an ANN, it would be quite challenging to locate all of the appropriate weights by hand. Thankfully, we can utilize algorithms to automatically modify the weights such that the network, after learning from its experience or being trained, generates the output we want.

One of the algorithms that is utilized in the training of neural networks is called backpropagation. It is a form of supervised learning, in which the intended output (target) is known in advance, and the network adjusts its weights and biases in order to create the desired output. In this type of learning, the desired output is produced by the network by adjusting its weights and biases. The error is “propagated” (or “flown”) from the output layer via the hidden levels until it reaches the input layer in the process of backpropagation. Then, both the weights and the biases are modified in order to reduce the amount of error. In practice, gradient descent is frequently used in combination with backpropagation. This optimization method is used to locate the group of weights that results in the lowest possible cost function, which is a measure of how much the projected output deviates from the actual output.

Neuronal networks have the ability to learn associations that are not linear. This is because there are several hidden layers in between the input and output layers, which enables more sophisticated mappings of input to output. The activation function keeps each layer separate: linear *times* linear = linear. This is the reason why there are multiple hidden levels. On the other hand, linear models, such as logistic regression, are only able to discover linear correlations in data. Neural networks provide a number of benefits, one of which is that they are quite versatile in terms of the kinds of data with which they are able to operate. They are able to process both numerical and categorical data, which lends to their versatility as a data processing method. In addition, when working with vast volumes of data, neural networks have a tendency to provide more accurate results when compared to other machine learning approaches. Finally, after a neural network has been trained on a dataset, it can be used to generate predictions on extremely fast on fresh data. This is because the network has learned how to best interpret the original data.

There are however certain drawbacks associated with the utilization of neural networks. One of the most significant negatives is that they may be quite challenging

to train, and it may require a significant amount of time and effort before they begin to perform well. In addition, because neural networks are so adaptable, they have the potential to sometimes produce inaccurate results if they are not employed with extreme caution, e.g. overfitting (described above). This indicates that their accuracy on the training data may be high, but their performance on newly acquired data may be subpar. As a last point of consideration, neural networks call for a significant amount of processing capacity, which might render their utilization in some contexts problematic.

6.3 Physics-Informed Neural Networks

A physics-informed neural network (PINN) [37] is a neural network that is trained to solve problems in physics by learning from data and equations that describe physical phenomena. PINNs can be used to solve problems in a wide range of fields, including mechanics, electromagnetism, and thermodynamics. PINNs are based on the concept of constraining neural networks with partial differential equations (PDEs). The PDEs serve as regularizers, avoiding overfitting and guaranteeing that the learned function respects the governing physics. In addition, the PDEs can lead the search for optimum solutions in high-dimensional spaces. PINNs may be trained using gradient-based optimization techniques such as ADAM or LBFGS. The primary benefit of PINNs is that they may be utilized to solve PDEs without discretization. This is in contrast to conventional numerical approaches, which require discretization of the PDEs prior to solution.

Constructing a PINN is straightforward. Let us denote the PDE as [37]

$$f(x, t) \equiv \mathcal{N}[u, x, t] = 0, x \in \Omega \quad (6.1)$$

where $\mathcal{N}[\cdot]$ is a differential operator, u is the solution of the PDE and Ω is some domain in $\mathbb{R}^d$ and $x = [x_1, x_2, \dots, x_d]$. Then, we define a neural network to approximate the

solution for Eq.6.1 and its derivatives with respect to the inputs (x and t). This neural network structure is illustrated in Fig. 6.2.

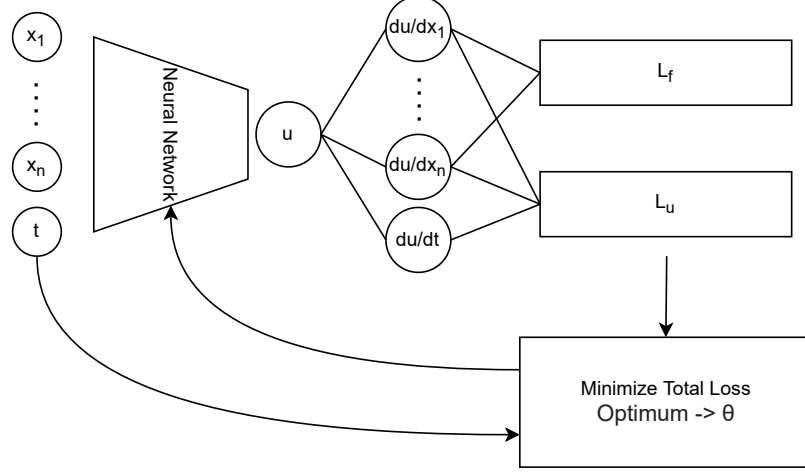


Figure 6.2: Network structure of a PINN.

The network is trained by minimizing a loss function defined as

$$L(\theta) = L_f(\theta) + L_u(\theta). \quad (6.2)$$

The first term in Eq. 6.2 is the loss contribution from the PDE 6.1 which is defined by

$$L_f(\theta) = \frac{1}{N_c} \sum_{i=1}^{N_c} |f(x_c^i, t_c^i)|^2 \quad (6.3)$$

where f is evaluated at the collocation points in the domain. The collocation points are denoted by the subscript c and N_c is the number of these points. Calculation of the loss Eq. 6.3 involves the differentiation of the output function u with respect to x and t . This is achieved by the automatic differentiation technique [26]. For large tasks, symbolic differentiation can be sluggish and inefficient. Inaccurate numerical differentiation might occur if the step size is not properly selected. Automatic differentiation is often more precise and quicker than symbolic differentiation. The second term in Eq. 6.2 is the loss contribution from initial and boundary training data. It

is given by

$$L_u(\theta) = \frac{1}{N_u} \sum_{i=1}^{N_u} |u(x_u^i, t_u^i) - u^i|^2 \quad (6.4)$$

where u^i is the training data at (x_u^i, t_u^i) .

The fact that PINNs may be trained with a very small number of data points is one of its primary advantages. This is possible due to the fact that the neural network is able to learn from the training data while being regularized by the PDE loss Eq. 6.3. This indicates that the neural network is capable of making correct predictions in places where no training data exists. This is especially useful for solving PDEs with complicated geometries.

6.4 Magnetic Field Mapping of Inaccessible Regions Using Physics-Informed Neural Networks

6.4.1 Methods

In this work, we are interested in predicting the magnetic field components inside a three dimensional space enclosed by an external surface S by utilizing the knowledge of the magnetic field in some number of locations on the surface S . Assuming there are no free currents, $\vec{J} = 0$, and no magnetization, $\vec{M} = 0$, in the region of interest, the partial differential equations that govern the static magnetic field are quite concise. They are

$$\vec{\nabla} \cdot \vec{B} = 0 \quad (6.5)$$

and

$$\vec{\nabla} \times \vec{B} = 0. \quad (6.6)$$

Therefore, it is possible to find a magnetic field that satisfies (6.5) and (6.6) together with the knowledge of the magnetic field at some number of locations . We choose those locations on the surface of a closed region S of which we are interested in approximating the magnetic field inside (Fig. 6.4). Of course, according to the

electromagnetism uniqueness theorem, having a finite number of data on a surface does not guarantee a unique solution to equations (6.5) and (6.6). However, as we demonstrate later in this paper, our results indicate that one can successfully approximate the true solution by having a sufficient number of data scattered on the surface.

Review of the Multipole Expansion Method

In this section, we review the field monitoring method described in [32]. Eq. (6.6) indicates that the magnetic field vector can be written as a gradient of a scalar magnetic potential function

$$\vec{B} = -\nabla\Phi_M(\vec{r}). \quad (6.7)$$

Substituting (6.7) into Eq. (6.5) tells us that the magnetic scalar potential satisfies Laplace's equation

$$\nabla^2\Phi_M = 0 \quad (6.8)$$

and the solution of Laplace's equation in spherical coordinates is given by

$$\Phi_M(r, \theta, \phi) = \sum_l^\infty \sum_m^l r^l P_l^m(\cos \theta) [a_{lm} \cos(m\phi) + b_{lm} \sin(m\phi)] \quad (6.9)$$

where P_l^m are the associated Legendre polynomials, and a_{lm} and b_{lm} are expansion coefficients. The magnetic field can be obtained by calculating the gradient of the magnetic scalar potential, $\vec{B} = -\nabla\Phi_M(\vec{r})$. Absorbing a_{lm} and b_{lm} in a coefficient c_n , we can write the magnetic field in a compact form

$$\vec{B}(x, y, z) = \sum_n c_n \vec{f}_n(x, y, z) \quad (6.10)$$

where $\vec{f}_n(x, y, z)$ are vector basis functions satisfying $\nabla \cdot \vec{f}_n = 0$ and $\nabla \times \vec{f}_n = 0$. To illustrate, the first 10 $\vec{f}_n$ basis vector functions are listed in Table 6.1. The right-hand side of the Eq. (6.10) is expanded to some finite order $n = N$ and the magnetic field vector inside the volume can be interpolated using linear regression techniques.

Table 6.1: Table of the x, y and z components of the basis vector function $\vec{f}_n$ up to order $n = 9$

	$\vec{f}_0$	$\vec{f}_1$	$\vec{f}_2$	$\vec{f}_3$	$\vec{f}_4$	$\vec{f}_5$	$\vec{f}_6$	$\vec{f}_7$	$\vec{f}_8$	$\vec{f}_9$
x	0	-1	0	$-x$	$-3z$	0	$6x$	$6y$	$-3xz$	$\frac{3}{2}(3x^2 + y^2 - 4z^2)$
y	0	0	-1	$-y$	0	$-3z$	$-6y$	$6x$	$-3yz$	$3xy$
z	1	0	0	$2z$	$-3x$	$-3y$	0	0	$-\frac{3}{2}(x^2 + y^2 - 2z^2)$	$-12xz$

Magnetic Field Prediction Using PINNs

The exact values of the partial derivatives in (6.5) and (6.6) can be calculated by automatic differentiation [37], which is implemented in some well-known machine learning libraries such as TensorFlow [1] and PyTorch [34]. The neural network we train to approximate the magnetic field inside the region will have the structure as shown in Fig. 6.3. The hyperbolic tangent is used for the activation of each hidden layer. The other tested activation functions have not performed as well as the hyperbolic tangent for this network architecture. The number of hidden layers are chosen to be 4 and 8 with each having 32 or 64 neurons. The performance of these 4 different-sized networks are discussed later. Then, the network can be trained by a

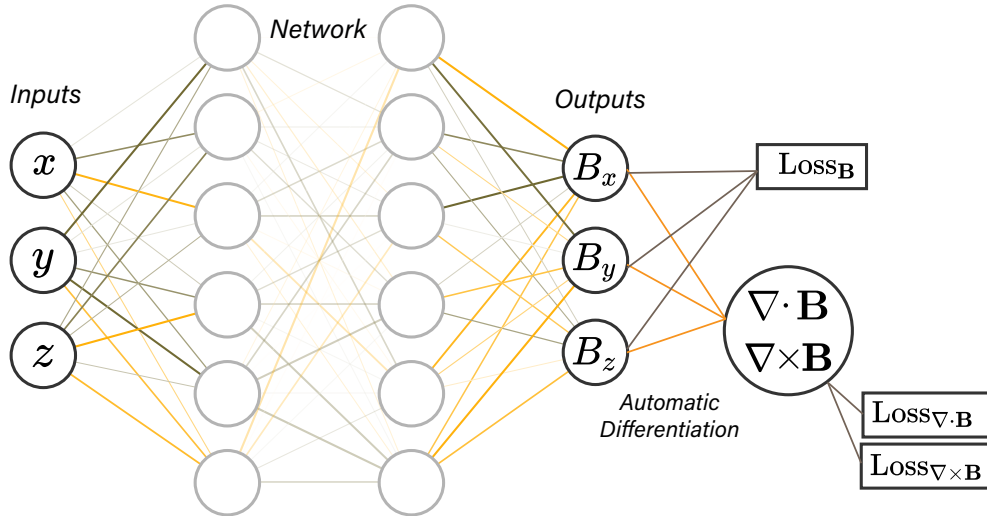


Figure 6.3: Network takes 3 inputs, (x, y, z) coordinates, and outputs the magnetic field $\vec{B}$. Automatic differentiation is used to calculate the exact derivatives of the output $\vec{B}$ with respect to the input parameters.

combined loss function of data, curl and divergence losses

$$\text{Loss} = \text{Loss}_{\vec{B}} + \lambda(\text{Loss}_{\vec{\nabla} \cdot \vec{B}} + \text{Loss}_{\vec{\nabla} \times \vec{B}}) \quad (6.11)$$

where

$$\text{Loss}_{\vec{B}} := \frac{1}{N_{\vec{B}}} \sum_{i=1}^{N_{\vec{B}}} \|\vec{B}(\vec{r}_{\vec{B}}^i) - \vec{B}_s(\vec{r}_{\vec{B}}^i)\|^2, \quad (6.12)$$

$$\text{Loss}_{\vec{\nabla} \cdot \vec{B}} := \frac{1}{N_f} \sum_{i=1}^{N_f} |\vec{\nabla} \cdot \vec{B}(\vec{r}_d^i)|^2 \quad (6.13)$$

and

$$\text{Loss}_{\vec{\nabla} \times \vec{B}} := \frac{1}{N_f} \sum_{i=1}^{N_f} \|\vec{\nabla} \times \vec{B}(\vec{r}_d^i)\|^2 \quad (6.14)$$

where the points $\vec{r}_{\vec{B}}^i$ and $\vec{r}_d^i$ denote the positions of the magnetic sensors and the collocation points, respectively. $N_{\vec{B}}$ is the number of the magnetic field sensors, N_f is the number of collocation points in the domain and $\vec{B}_s$ is the measured magnetic field vector at $\vec{r}_d^i$. The parameter λ in Eq. (6.11) can be adjusted according to the performance of the network. The collocation points, $\vec{r}_d^i$, in equations (6.13) and (6.14) are sampled from the volume encapsulated by the surface S (Fig. 6.4) and can be chosen to be fixed throughout the training process [37]. However, randomly choos-

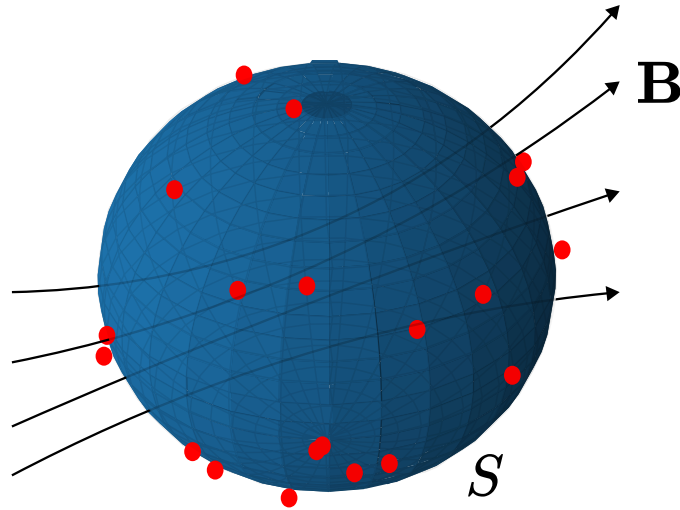


Figure 6.4: Magnetic field sensors (red dots) placed on a surface S to predict the magnetic field $\vec{B}$ in the inner region

ing collocation points in each epoch leads to a quicker convergence as well as more accurate results. This is partly due to being able to choose fewer number of collocation points, and since they are assigned randomly each iteration, they represent the domain better than any fixed collocation points scheme. The ADAM optimizer [24], which is an adaptive method for gradient-based first-order optimization, is what we make use of in order to minimize the loss function 6.11. The general procedure for training is given in Algorithm 1.

```

 $\vec{r}_{\vec{B}}^i \leftarrow$  sensor locations;
 $\vec{B}_s(\vec{r}_{\vec{B}}^i) \leftarrow$  sensor data;
while  $Loss \geq \epsilon$  do
     $\vec{r}_d^i \leftarrow$  random locations inside the region;
    Compute  $Loss_{\vec{B}}$  by (6.12);
    Compute  $Loss_{\vec{\nabla} \cdot \vec{B}}$  by (6.13);
    Compute  $Loss_{\vec{\nabla} \times \vec{B}}$  by (6.14);
     $Loss \leftarrow Loss_{\vec{B}} + \lambda(Loss_{\vec{\nabla} \cdot \vec{B}} + Loss_{\vec{\nabla} \times \vec{B}})$ ;
    update the weights of the network according to  $Loss$ ;
end

```

Algorithm 1: Training algorithm of the network

6.4.2 Experiments

Simulated Experiment

In the following example, we will demonstrate the capability of our magnetic field prediction model by placing an arbitrary number of triple-axis magnetic sensors on the surface of a cube. The model will be trained and validated by using the analytical expression of magnetic field vector for a circular current loop. Then we will use the trained model to predict the direction of magnetic flux density vector on arbitrary points inside or outside of cube. In this section, we are going to demonstrate how our three dimensional prediction method works for multiple triple-axis sensors placed randomly on different surfaces(e.g., faces) of a cube. Magnetic field sensors will be

placed on the cube randomly and we will generate training and validation data by using Biot-Savart law for circular current loop(s). We will use magnetic field data to train our neural network and then we will use the trained model to predict the direction of magnetic flux density vector on arbitrary points. In the next section, we will give the analytical expression of the three dimensional magnetic field vector of a single circular current loop and then we will construct a higher order asymmetric magnetic field by placing multiple loops with different currents to benchmark our magnetic field prediction model.

We begin by demonstrating the ability of our magnetic field reconstruction method by considering the magnetic field of a simple circular current loop (in arbitrary units). The magnetic field components of a circular current loop with radius a are given by [21, 9]

$$\begin{aligned} B_x &= C \frac{xz}{2\alpha^2\rho^2\beta} [(a^2 + r^2)E(k^2) - \alpha^2 K(k^2)] \\ B_y &= \frac{y}{x} B_x \\ B_z &= C \frac{1}{2\alpha^2\beta} [(a^2 - r^2)E(k^2) + \alpha^2 K(k^2)] \end{aligned} \tag{6.15}$$

with

$$k^2 = \frac{4ar \sin \theta}{a^2 + r^2 + 2ar \sin \theta}$$

where $E(k)$ and $K(k)$ are elliptic integrals, $\rho^2 \equiv x^2 + y^2$, $\alpha^2 \equiv a^2 + r^2 - 2a\rho$, $\beta^2 \equiv a^2 + r^2 + 2a\rho$ and $r \equiv \sqrt{x^2 + y^2 + z^2}$ and $z = r \cos \theta$. In this work, we will work in arbitrary units by setting $C = 1$.

We want to show the potential of the network by comparing it to the multipole expansion method for various sensor counts and different types and levels of noises. To create a non-uniform higher order magnetic field, we positioned 8 circular loops of different current values at positions $(x = \pm 1.01, y = \pm 1, z = \pm 4)$ and the triple-axis magnetic sensors are placed randomly on the surface of a cube with side length $L = 2$

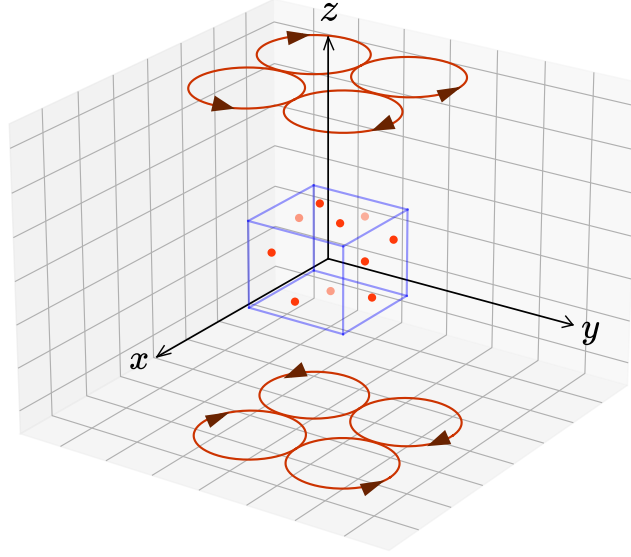


Figure 6.5: Configuration of the test model. Red circles: current loops, blue cube: cubical sensor array and red dot: triple-axis magnetic field sensors.

centered at the origin. The configuration is illustrated in Fig. 6.5. Our goal is to predict the magnetic field in the inner region of the surface.

The number of hidden layers and neurons of the network characterizes the complexity of the function it can approximate. Having more hidden layers and neurons should not negatively affect the performance, training larger networks are slower and may require more care with initialization and regularization of the weights [20]. In this example, larger network sizes resulted in better performance as shown in Table 6.2 as expected. Models were trained for less than 2 minutes for all cases on an NVIDIA RTX 3080 GPU.

Table 6.2: Error between the predicted and exact magnetic field in vector norm for various sensor counts and network structures.

$N_{\vec{B}}$	Network Size	4×32	4×64	8×32	8×64
18		1.9×10^{-2}	2.4×10^{-2}	2.2×10^{-2}	2.2×10^{-2}
30		8.5×10^{-3}	8.2×10^{-3}	7.9×10^{-3}	7.9×10^{-3}
60		1.9×10^{-3}	1.9×10^{-3}	1.9×10^{-3}	1.9×10^{-3}
90		1.7×10^{-3}	1.7×10^{-3}	1.7×10^{-3}	1.7×10^{-3}

When more sensors are added to a system, more information is gathered about it.

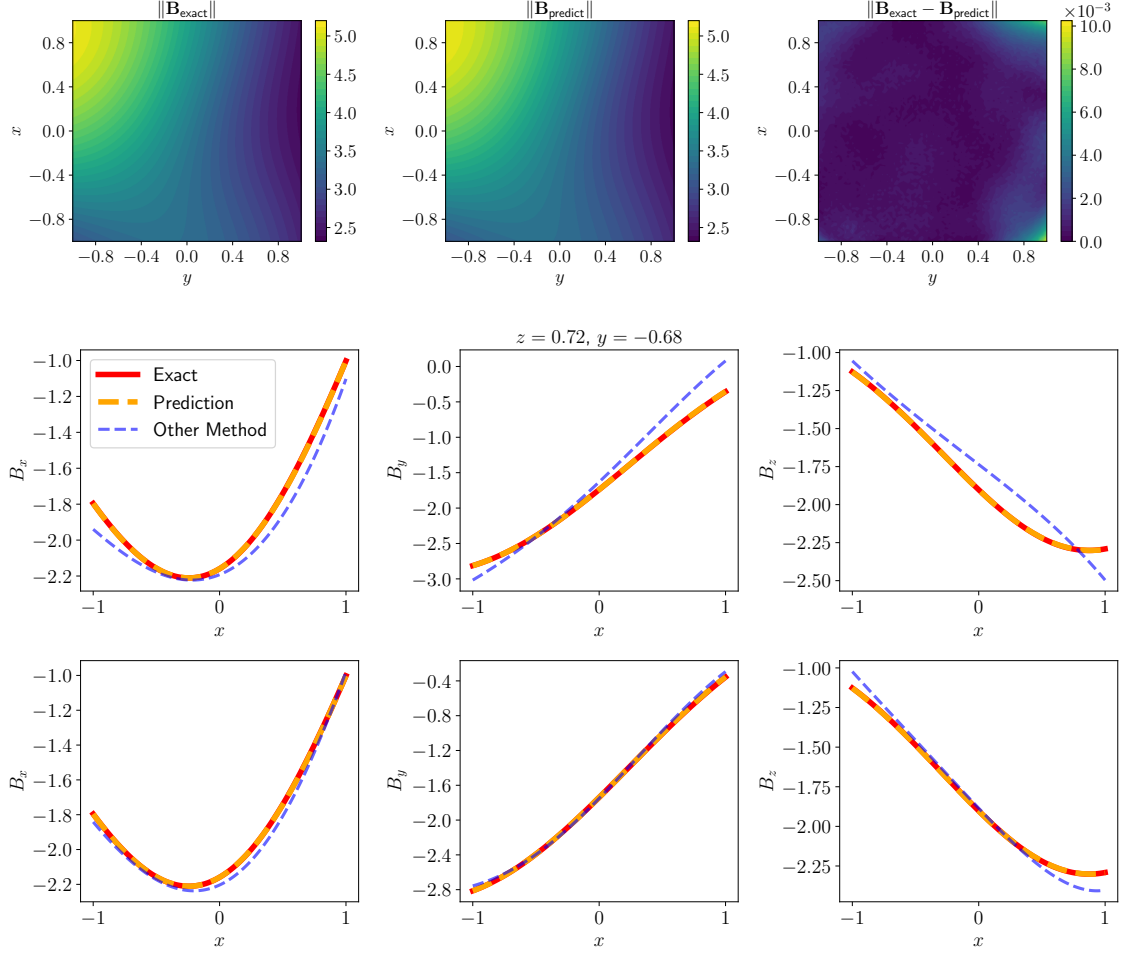


Figure 6.6: Top: Magnitude plot of the predicted magnetic field along with exact magnetic field and the error at snapshot $z = 0.72$, Middle: Comparison of the predicted and the other method's solutions' x, y, z directions with 18 sensor data, Bottom: Comparison of the predicted and the other method's solutions' x, y, z directions with 30 sensor data.

Machine learning models tend to improve their predictive abilities with increased data. For this reason, greater sensor counts gives more information about the magnetic field of the system, and we would expect the network to be able to use that information to predict magnetic field better. As shown in Table 6.2, having more sensory information has led to better performance for all network structures. Moreover, fewer sensor counts has not led to a divergence from the exact magnetic field. This is not the case with the multipole expansion method as shown in Table 6.3. The other method seems

Table 6.3: Error measure in vector norm for the predicted and the other method's different orders for various sensor counts.

Sensor Count	PINN	4th Order	6th Order	8th Order
18	2.4×10^{-2}	2.0×10^{-1}	8.1	25
30	8.2×10^{-3}	1.5×10^{-1}	2.7×10^{-2}	10
60	2.9×10^{-3}	1.3×10^{-1}	1.6×10^{-2}	2.0×10^{-3}
90	1.8×10^{-3}	1.3×10^{-1}	1.4×10^{-2}	2.0×10^{-3}

to suffer with relatively few sensors and higher order versions overfit the sensory data. Decreasing the order in this case leads to better results but due to lower orders having fewer basis functions, the method is not able to predict the exact magnetic field as well as our network. This can also be seen in Fig. 6.6 and 6.7.

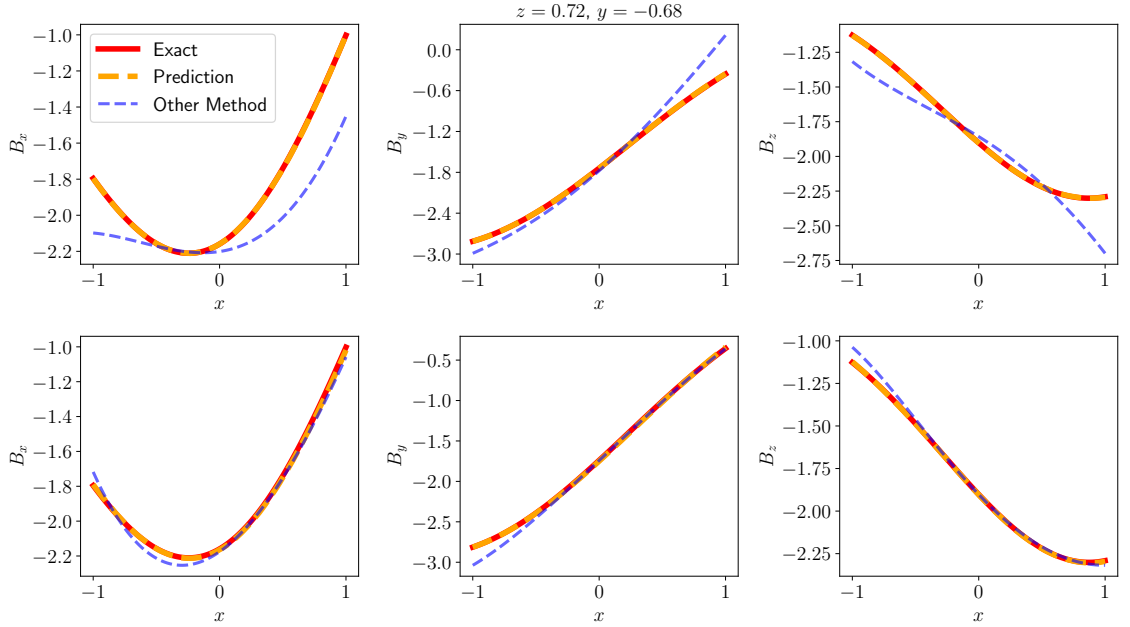


Figure 6.7: Top: Comparison of the predicted and the other method's solutions' x, y, z directions with 18 sensor data with Gaussian noise with $\sigma = 1.0 \times 10^{-2}$, Bottom: Comparison of the predicted and the other method's solutions' x, y, z directions with 30 sensor data with Gaussian noise with $\sigma = 1.0 \times 10^{-2}$.

Performance of the network when Gaussian noise is introduced to the sensory information is given in Table 6.4. This noise has led to a further deterioration of the performance for the multipole expansion method. Our method has also been affected, however, performed better across various sensor counts.

Table 6.4: Error measure in vector norm for the predicted and the other method for various sensor counts with Gaussian noise added to the sensor data with standard deviations $\sigma = 5.0 \times 10^{-3}$ and $\sigma = 1.0 \times 10^{-2}$.

Sensor Count	PINN (with noise)		Other Method (with noise)	
	5.0×10^{-3}	1.0×10^{-2}	5.0×10^{-3}	1.0×10^{-2}
18	2.6×10^{-2}	4.1×10^{-2}	2.0×10^{-1}	2.1×10^{-1}
30	1.2×10^{-2}	1.5×10^{-2}	2.9×10^{-2}	4.0×10^{-2}
60	5.0×10^{-3}	6.2×10^{-3}	8.0×10^{-3}	1.6×10^{-2}
90	4.5×10^{-3}	5.6×10^{-3}	8.0×10^{-3}	1.6×10^{-2}

Mapping the Magnetic Field of a Square Coil System

To demonstrate our methodology using actual data, we conducted an experiment in which a Bartington triple axis magnetic field probe (Mag-13MS1000) was moved to the locations of the training data collection points. In order to generate a non-uniform magnetic field, two rectangular coils are stacked vertically and driven with different current magnitudes in the opposite directions (Fig. 6.8). The resulting field profile is non-uniform and depends on the coil configuration and current magnitudes. Each face of the coils is a printed circuit board (PCB) with dimensions $55 \text{ cm} \times 16 \text{ cm}$ containing 50 parallel line traces along the long side of the PCB. A current with magnitude 1 A flows counterclockwise and another current with magnitude 0.6 A flows clockwise in the top and bottom coils respectively.

In order to isolate the field generated by the coils, at each measurement location, the data is collected as the difference between the sensor measurements with the coil turned on and off. Then we train the network for the magnetic field data collected using the magnetic mapping system. The training domain is chosen as a cube with 40 cm side length centered at the center of the coils. Performance benchmarks of our network and the other method on the measurement data is given in Fig. 6.9 and Fig. 6.10.

6.4.3 Conclusions

In this study, we presented an efficient and practical method for mapping the magnetic field of inaccessible locations. We encoded previous knowledge from Maxwell's equations for magnetostatics into a physics-informed neural network model for magnetic field prediction in regions where direct measurements are not possible.

We provided two experiments that proved the practicability of the proposed method. A simulated experiment proved the value of incorporating extra physics knowledge into the model. Mapping the magnetic field of a square coil system illustrated the effectiveness of the approximation technique in real world applications. We were able to predict the strength of the magnetic field at any point surrounding a coil system by using the trained model. Finally, we showed that our technique may be utilized for online prediction of magnetic fields.

The performance of the physics-informed neural network model was compared with that of the multipole expansion method, which uses analytical solutions for magnetic fields. The results showed that our method outperformed traditional methods in its ability to learn complex nonlinear relationships and map them to a new set of data such as measuring locations where there are no direct measurements. The comparison of our method and multipole expansion method indicated better performance results across various sensor counts and noise levels both in simulation data and real world measurement data.

Because it is based on a data-driven approach, the method that has been described has the potential benefit of not requiring any analytical solutions for magnetic fields to be known in advance. This is a significant advantage. The trained model is able to directly map the magnetic field of an arbitrary coil system.

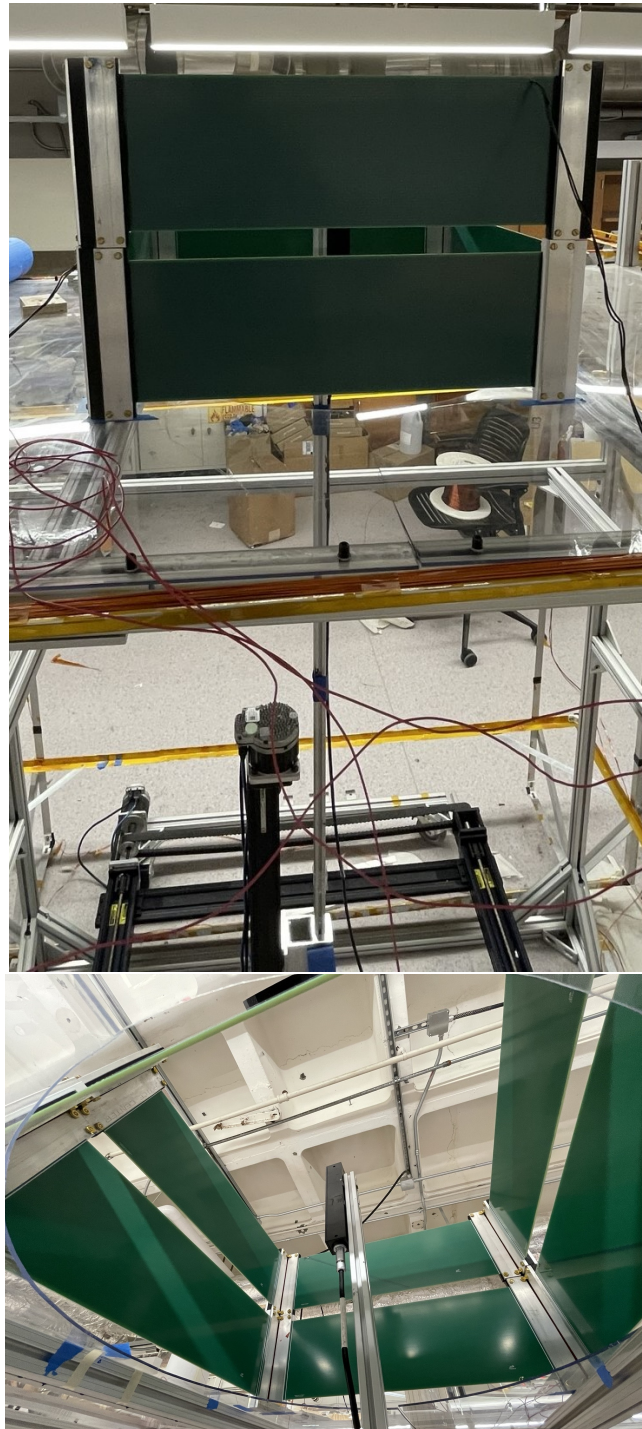


Figure 6.8: Setup of the experiment. Top: 3D magnetic field mapping system is sitting underneath the rectangular coils. The bottom figure shows the triple-axis magnetometer and inside the coils.

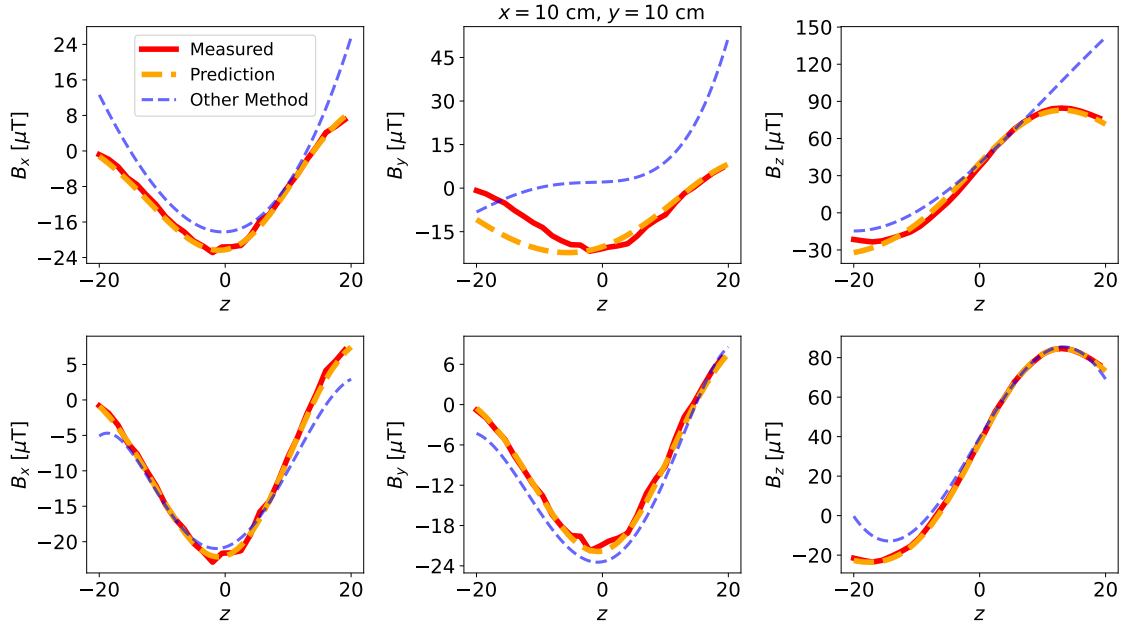


Figure 6.9: Comparison of the predicted and the other method's solutions along $x = 10$, $y = 10$ axis. Top: 18 sensor data, Bottom: 30 sensor data.

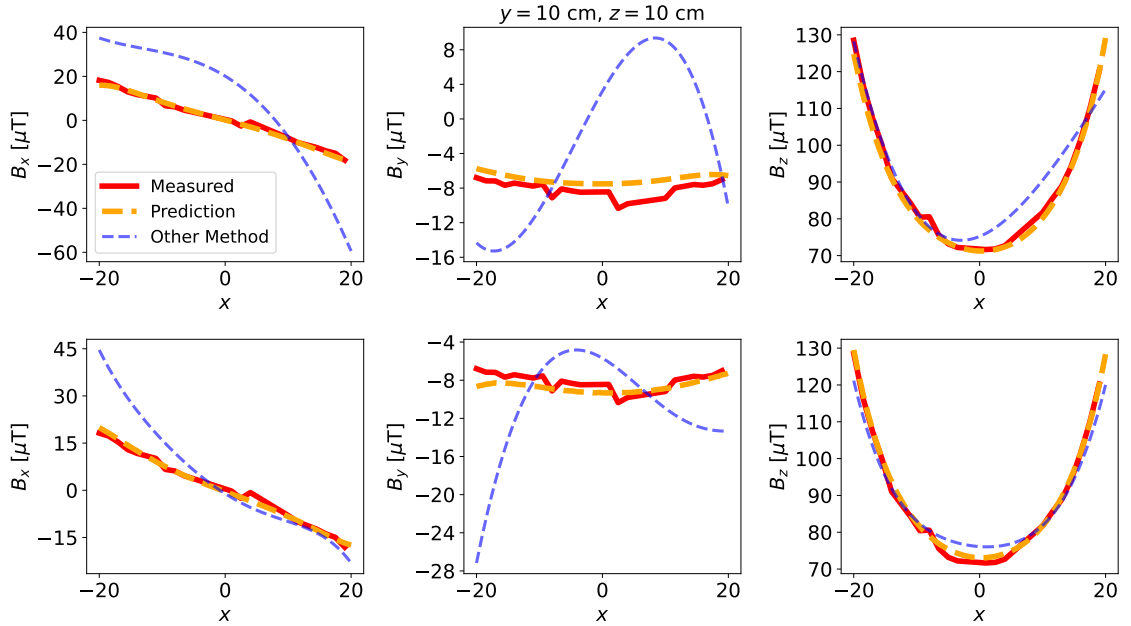


Figure 6.10: Comparison of the predicted and the other method's solutions along $y = 10$, $z = 10$ axis. Top: 18 sensor data, Bottom: 30 sensor data.

Appendices

Appendix A: Arduino Firmware Code for the Probe Switching Box

```
int incomingByte = 0;

byte nullByte = 0b00000000;
byte toggByte = 0b00000001;

byte byteList[39][6];

for(int i=0; i<6; i++){
    for(int j=0; j<39; j++){
        if(i==5 - j/8){
            byteList[j][i] = toggByte << ((j + j/8) % 8);
        }
        else{
            byteList[j][i] = nullByte;
        }
    }
}

// define pins and other variables
const int latchPin = 3;
const int clockPin = 2;
const int dataPin = 5;
const int oePin = 4;
```



```

bool hasRun = false;

void outActivate(unsigned int outNumber)
{
    digitalWrite(latchPin, LOW);
    digitalWrite(oePin, HIGH);
    for(int i=0; i<6; i++)
    {
        shiftOut(dataPin, clockPin, MSBFIRST, byteList[outNumber][i]);
    }
    digitalWrite(oePin, LOW);
    digitalWrite(latchPin, HIGH);
}

void setup()
{
    Serial.begin(9600);
    delay(100);
    Serial.println("<Probe Switcher is Ready>");

    //set pins to output because they are addressed in the main loop
    pinMode(latchPin, OUTPUT);
    pinMode(clockPin, OUTPUT);
    pinMode(dataPin, OUTPUT);
    pinMode(oePin, OUTPUT);
    digitalWrite(oePin, LOW);

```

```

}

void loop()
{
  if(Serial.available() > 1)
  {
    int v1 = Serial.read();
    int v2 = Serial.read();

    int activatedNumber = (v1 - 48)*10 + (v2 - 48);
    outActivate(activatedNumber);
    Serial.print(activatedNumber);

    Serial.println(" is activated");
  }
}

```

Appendix B: Field Monitor Test Results

Table B1: Measurement values for $I = 50$ mA

x (cm)	y (cm)	z (cm)	B_x (mG)	Reconstruction			B_x (mG)	Measured	
				B_y (mG)	B_z (mG)	B_z (mG)		B_y (mG)	B_z (mG)
-16.00	0.00	0.00	92.07	166.24	-415.95	92.21	166.41	-415.86	
-14.00	0.00	0.00	92.17	166.19	-415.83	92.40	166.37	-415.70	
-12.00	0.00	0.00	92.26	166.15	-415.71	92.45	166.30	-415.67	
-10.00	0.00	0.00	92.36	166.11	-415.60	92.53	166.26	-415.54	
-8.00	0.00	0.00	92.47	166.07	-415.48	92.60	166.20	-415.44	
-6.00	0.00	0.00	92.57	166.04	-415.36	92.67	166.12	-415.36	
-4.00	0.00	0.00	92.68	166.01	-415.24	92.73	166.06	-415.29	
-2.00	0.00	0.00	92.80	165.98	-415.13	92.82	165.98	-415.18	
0.00	0.00	0.00	92.91	165.95	-415.01	92.88	165.87	-415.10	
2.00	0.00	0.00	93.03	165.92	-414.89	92.98	165.82	-415.00	
4.00	0.00	0.00	93.15	165.89	-414.77	93.07	165.74	-414.90	
6.00	0.00	0.00	93.27	165.86	-414.65	93.17	165.67	-414.79	
8.00	0.00	0.00	93.39	165.82	-414.53	93.28	165.59	-414.69	
10.00	0.00	0.00	93.51	165.78	-414.42	93.38	165.52	-414.62	
12.00	0.00	0.00	93.63	165.74	-414.30	93.49	165.44	-414.49	
14.00	0.00	0.00	93.74	165.69	-414.18	93.60	165.37	-414.36	
16.00	0.00	0.00	93.86	165.64	-414.07	93.72	165.26	-414.28	
0.00	-20.00	0.00	92.96	167.73	-415.63	93.08	167.91	-415.68	
0.00	-18.00	0.00	92.99	167.54	-415.58	93.12	167.73	-415.64	
0.00	-16.00	0.00	93.01	167.35	-415.53	93.11	167.56	-415.58	
0.00	-14.00	0.00	93.02	167.17	-415.48	93.10	167.34	-415.52	
0.00	-12.00	0.00	93.02	166.99	-415.42	93.07	167.16	-415.47	
0.00	-10.00	0.00	93.02	166.82	-415.36	93.01	166.96	-415.41	
0.00	-8.00	0.00	93.01	166.64	-415.29	93.02	166.77	-415.33	
0.00	-6.00	0.00	92.99	166.47	-415.22	93.00	166.59	-415.25	
0.00	-4.00	0.00	92.97	166.30	-415.15	92.97	166.37	-415.23	
0.00	-2.00	0.00	92.94	166.13	-415.08	92.95	166.20	-415.13	
0.00	0.00	0.00	92.91	165.95	-415.01	92.94	166.00	-415.05	
0.00	2.00	0.00	92.88	165.78	-414.93	92.92	165.79	-414.96	
0.00	4.00	0.00	92.85	165.60	-414.86	92.91	165.59	-414.88	
0.00	6.00	0.00	92.82	165.42	-414.79	92.87	165.37	-414.77	
0.00	8.00	0.00	92.78	165.23	-414.72	92.84	165.13	-414.73	
0.00	10.00	0.00	92.75	165.04	-414.66	92.80	164.95	-414.66	
0.00	12.00	0.00	92.71	164.85	-414.59	92.77	164.69	-414.59	
0.00	14.00	0.00	92.68	164.65	-414.53	92.73	164.51	-414.49	
0.00	16.00	0.00	92.64	164.44	-414.48	92.72	164.29	-414.34	
0.00	18.00	0.00	92.60	164.23	-414.42	92.67	164.14	-414.27	
0.00	20.00	0.00	92.56	164.00	-414.37	92.58	163.97	-414.18	
0.00	0.00	-8.00	92.43	165.68	-415.24	92.27	165.62	-415.45	
0.00	0.00	-6.00	92.55	165.75	-415.18	92.58	165.66	-415.34	
0.00	0.00	-4.00	92.67	165.81	-415.12	92.65	165.78	-415.25	
0.00	0.00	-2.00	92.79	165.88	-415.06	92.82	165.83	-415.15	
0.00	0.00	0.00	92.91	165.95	-415.01	92.90	165.94	-415.06	
0.00	0.00	2.00	93.03	166.03	-414.95	93.09	165.97	-414.95	
0.00	0.00	4.00	93.15	166.10	-414.89	93.14	166.07	-414.90	
0.00	0.00	6.00	93.27	166.18	-414.83	93.35	166.12	-414.78	
0.00	0.00	8.00	93.39	166.26	-414.77	93.37	166.19	-414.71	

Table B2: Measurement values for $I = 100$ mA

x (cm)	y (cm)	z (cm)	B_x (mG)	Reconstruction			B_x (mG)	Measured	
				B_y (mG)	B_z (mG)	B_z (mG)		B_y (mG)	B_z (mG)
-16.00	0.00	0.00	91.76	166.25	-415.92	92.04	166.44	-415.83	
-14.00	0.00	0.00	91.89	166.20	-415.81	92.13	166.41	-415.74	
-12.00	0.00	0.00	92.03	166.16	-415.69	92.23	166.38	-415.58	
-10.00	0.00	0.00	92.16	166.12	-415.58	92.32	166.31	-415.50	
-8.00	0.00	0.00	92.30	166.08	-415.46	92.43	166.25	-415.42	
-6.00	0.00	0.00	92.45	166.05	-415.34	92.54	166.17	-415.36	
-4.00	0.00	0.00	92.60	166.02	-415.23	92.65	166.08	-415.24	
-2.00	0.00	0.00	92.75	165.99	-415.11	92.77	166.02	-415.14	
0.00	0.00	0.00	92.90	165.96	-414.99	92.88	165.94	-415.05	
2.00	0.00	0.00	93.06	165.94	-414.87	93.05	165.89	-414.96	
4.00	0.00	0.00	93.21	165.91	-414.76	93.13	165.79	-414.86	
6.00	0.00	0.00	93.37	165.87	-414.64	93.26	165.73	-414.76	
8.00	0.00	0.00	93.53	165.84	-414.52	93.39	165.63	-414.61	
10.00	0.00	0.00	93.69	165.80	-414.40	93.54	165.57	-414.56	
12.00	0.00	0.00	93.85	165.76	-414.29	93.68	165.49	-414.49	
14.00	0.00	0.00	94.00	165.71	-414.17	93.83	165.40	-414.37	
16.00	0.00	0.00	94.16	165.66	-414.06	93.92	165.33	-414.24	
0.00	-20.00	0.00	92.92	167.98	-415.60	93.03	168.10	-415.67	
0.00	-18.00	0.00	92.96	167.76	-415.56	93.10	167.92	-415.61	
0.00	-16.00	0.00	92.98	167.55	-415.51	93.08	167.72	-415.53	
0.00	-14.00	0.00	92.99	167.34	-415.46	93.00	167.49	-415.51	
0.00	-12.00	0.00	93.00	167.14	-415.40	93.06	167.28	-415.48	
0.00	-10.00	0.00	93.00	166.94	-415.34	93.03	167.07	-415.36	
0.00	-8.00	0.00	92.99	166.74	-415.27	93.03	166.86	-415.34	
0.00	-6.00	0.00	92.97	166.55	-415.20	92.98	166.66	-415.24	
0.00	-4.00	0.00	92.95	166.35	-415.13	92.96	166.46	-415.18	
0.00	-2.00	0.00	92.93	166.16	-415.06	92.91	166.20	-415.09	
0.00	0.00	0.00	92.90	165.96	-414.99	92.94	166.00	-415.03	
0.00	2.00	0.00	92.87	165.77	-414.92	92.92	165.77	-414.94	
0.00	4.00	0.00	92.84	165.57	-414.85	92.89	165.55	-414.88	
0.00	6.00	0.00	92.81	165.37	-414.78	92.86	165.31	-414.79	
0.00	8.00	0.00	92.77	165.16	-414.71	92.81	165.06	-414.74	
0.00	10.00	0.00	92.74	164.95	-414.64	92.80	164.81	-414.61	
0.00	12.00	0.00	92.70	164.74	-414.58	92.76	164.60	-414.54	
0.00	14.00	0.00	92.67	164.52	-414.52	92.74	164.40	-414.44	
0.00	16.00	0.00	92.63	164.29	-414.46	92.69	164.19	-414.37	
0.00	18.00	0.00	92.59	164.06	-414.41	92.67	163.99	-414.30	
0.00	20.00	0.00	92.55	163.82	-414.36	92.65	163.80	-414.16	
0.00	0.00	-8.00	92.42	165.69	-415.16	92.26	165.66	-415.39	
0.00	0.00	-6.00	92.54	165.76	-415.11	92.58	165.70	-415.28	
0.00	0.00	-4.00	92.66	165.82	-415.07	92.64	165.78	-415.17	
0.00	0.00	-2.00	92.78	165.89	-415.03	92.84	165.87	-415.12	
0.00	0.00	0.00	92.90	165.96	-414.99	92.90	165.96	-415.02	
0.00	0.00	2.00	93.02	166.04	-414.95	93.08	166.01	-414.97	
0.00	0.00	4.00	93.14	166.11	-414.91	93.16	166.10	-414.91	
0.00	0.00	6.00	93.25	166.19	-414.87	93.34	166.13	-414.81	
0.00	0.00	8.00	93.37	166.27	-414.82	93.38	166.22	-414.80	

Table B3: Measurement values for $I = 150$ mA

x (cm)	y (cm)	z (cm)	B_x (mG)	Reconstruction			B_x (mG)	Measured	
				B_y (mG)	B_z (mG)	B_z (mG)		B_y (mG)	B_z (mG)
-16.00	0.00	0.00	91.46	166.26	-415.90	91.74	166.45	-415.81	
-14.00	0.00	0.00	91.63	166.21	-415.79	91.87	166.43	-415.68	
-12.00	0.00	0.00	91.80	166.17	-415.67	92.02	166.38	-415.62	
-10.00	0.00	0.00	91.97	166.13	-415.56	92.14	166.31	-415.51	
-8.00	0.00	0.00	92.15	166.09	-415.44	92.28	166.26	-415.42	
-6.00	0.00	0.00	92.33	166.06	-415.33	92.43	166.19	-415.34	
-4.00	0.00	0.00	92.51	166.03	-415.21	92.58	166.12	-415.23	
-2.00	0.00	0.00	92.70	166.00	-415.09	92.74	166.04	-415.15	
0.00	0.00	0.00	92.89	165.97	-414.98	92.91	165.97	-415.06	
2.00	0.00	0.00	93.08	165.95	-414.86	93.02	165.89	-414.93	
4.00	0.00	0.00	93.27	165.92	-414.75	93.19	165.83	-414.85	
6.00	0.00	0.00	93.46	165.88	-414.63	93.38	165.77	-414.75	
8.00	0.00	0.00	93.66	165.85	-414.51	93.54	165.68	-414.65	
10.00	0.00	0.00	93.85	165.81	-414.40	93.71	165.60	-414.54	
12.00	0.00	0.00	94.04	165.77	-414.29	93.88	165.51	-414.45	
14.00	0.00	0.00	94.23	165.72	-414.17	94.07	165.44	-414.32	
16.00	0.00	0.00	94.42	165.67	-414.06	94.27	165.35	-414.23	
0.00	-20.00	0.00	92.92	168.20	-415.59	93.02	168.25	-415.67	
0.00	-18.00	0.00	92.96	167.96	-415.55	93.09	168.09	-415.62	
0.00	-16.00	0.00	92.98	167.73	-415.50	93.03	167.86	-415.53	
0.00	-14.00	0.00	92.99	167.50	-415.45	93.05	167.66	-415.54	
0.00	-12.00	0.00	92.99	167.28	-415.39	93.03	167.41	-415.46	
0.00	-10.00	0.00	92.99	167.06	-415.33	93.01	167.17	-415.35	
0.00	-8.00	0.00	92.97	166.84	-415.26	92.98	166.95	-415.29	
0.00	-6.00	0.00	92.96	166.62	-415.19	92.92	166.72	-415.24	
0.00	-4.00	0.00	92.94	166.41	-415.12	92.94	166.50	-415.11	
0.00	-2.00	0.00	92.91	166.19	-415.05	92.93	166.25	-415.09	
0.00	0.00	0.00	92.89	165.97	-414.98	92.91	166.01	-414.92	
0.00	2.00	0.00	92.86	165.76	-414.91	92.89	165.78	-414.94	
0.00	4.00	0.00	92.83	165.54	-414.83	92.88	165.54	-414.84	
0.00	6.00	0.00	92.79	165.32	-414.76	92.84	165.25	-414.79	
0.00	8.00	0.00	92.76	165.09	-414.70	92.80	164.96	-414.72	
0.00	10.00	0.00	92.73	164.86	-414.63	92.79	164.74	-414.64	
0.00	12.00	0.00	92.69	164.63	-414.56	92.76	164.51	-414.53	
0.00	14.00	0.00	92.66	164.39	-414.50	92.72	164.27	-414.42	
0.00	16.00	0.00	92.62	164.14	-414.45	92.69	164.07	-414.36	
0.00	18.00	0.00	92.59	163.89	-414.39	92.66	163.85	-414.28	
0.00	20.00	0.00	92.55	163.63	-414.34	92.65	163.64	-414.20	
0.00	0.00	-8.00	92.42	165.70	-415.08	92.25	165.69	-415.33	
0.00	0.00	-6.00	92.54	165.76	-415.06	92.54	165.71	-415.22	
0.00	0.00	-4.00	92.65	165.83	-415.03	92.62	165.81	-415.18	
0.00	0.00	-2.00	92.77	165.90	-415.00	92.80	165.86	-415.08	
0.00	0.00	0.00	92.89	165.97	-414.98	92.87	165.97	-415.07	
0.00	0.00	2.00	93.00	166.05	-414.95	93.07	166.01	-414.98	
0.00	0.00	4.00	93.12	166.12	-414.92	93.12	166.11	-414.93	
0.00	0.00	6.00	93.24	166.20	-414.90	93.32	166.17	-414.90	
0.00	0.00	8.00	93.36	166.28	-414.87	93.37	166.22	-414.85	

Table B4: Measurement values for $I = 200$ mA

x (cm)	y (cm)	z (cm)	B_x (mG)	Reconstruction			B_x (mG)	Measured	
				B_y (mG)	B_z (mG)	B_z (mG)		B_y (mG)	B_z (mG)
-16.00	0.00	0.00	91.17	166.25	-415.93	91.46	166.46	-415.85	
-14.00	0.00	0.00	91.37	166.21	-415.81	91.62	166.43	-415.69	
-12.00	0.00	0.00	91.58	166.17	-415.69	91.80	166.38	-415.62	
-10.00	0.00	0.00	91.78	166.13	-415.57	91.96	166.33	-415.54	
-8.00	0.00	0.00	92.00	166.10	-415.46	92.13	166.25	-415.44	
-6.00	0.00	0.00	92.21	166.07	-415.34	92.30	166.19	-415.36	
-4.00	0.00	0.00	92.43	166.05	-415.22	92.48	166.12	-415.23	
-2.00	0.00	0.00	92.65	166.02	-415.10	92.67	166.04	-415.13	
0.00	0.00	0.00	92.87	166.00	-414.98	92.81	166.02	-415.02	
2.00	0.00	0.00	93.10	165.97	-414.86	93.04	165.92	-414.96	
4.00	0.00	0.00	93.33	165.95	-414.74	93.23	165.84	-414.87	
6.00	0.00	0.00	93.55	165.92	-414.62	93.44	165.78	-414.74	
8.00	0.00	0.00	93.78	165.89	-414.50	93.65	165.71	-414.62	
10.00	0.00	0.00	94.01	165.86	-414.39	93.86	165.63	-414.54	
12.00	0.00	0.00	94.24	165.82	-414.27	94.07	165.56	-414.43	
14.00	0.00	0.00	94.46	165.78	-414.15	94.33	165.47	-414.33	
16.00	0.00	0.00	94.69	165.73	-414.04	94.53	165.39	-414.23	
0.00	-20.00	0.00	92.87	168.43	-415.60	92.95	168.50	-415.65	
0.00	-18.00	0.00	92.90	168.16	-415.55	93.05	168.26	-415.62	
0.00	-16.00	0.00	92.93	167.91	-415.50	93.04	168.03	-415.56	
0.00	-14.00	0.00	92.95	167.66	-415.45	92.98	167.78	-415.50	
0.00	-12.00	0.00	92.95	167.41	-415.39	92.99	167.53	-415.26	
0.00	-10.00	0.00	92.95	167.17	-415.33	92.98	167.29	-415.37	
0.00	-8.00	0.00	92.94	166.94	-415.26	92.97	167.03	-415.30	
0.00	-6.00	0.00	92.93	166.70	-415.19	92.95	166.80	-415.21	
0.00	-4.00	0.00	92.92	166.47	-415.12	92.92	166.53	-415.12	
0.00	-2.00	0.00	92.89	166.23	-415.05	92.90	166.29	-415.11	
0.00	0.00	0.00	92.87	166.00	-414.98	92.89	166.02	-415.03	
0.00	2.00	0.00	92.85	165.76	-414.91	92.89	165.79	-414.94	
0.00	4.00	0.00	92.82	165.52	-414.84	92.84	165.52	-414.85	
0.00	6.00	0.00	92.79	165.29	-414.77	92.85	165.21	-414.77	
0.00	8.00	0.00	92.76	165.04	-414.70	92.82	164.94	-414.70	
0.00	10.00	0.00	92.73	164.80	-414.63	92.79	164.68	-414.69	
0.00	12.00	0.00	92.70	164.54	-414.57	92.76	164.42	-414.56	
0.00	14.00	0.00	92.67	164.29	-414.50	92.73	164.18	-414.45	
0.00	16.00	0.00	92.64	164.03	-414.45	92.70	163.95	-414.37	
0.00	18.00	0.00	92.60	163.76	-414.39	92.67	163.71	-414.29	
0.00	20.00	0.00	92.57	163.48	-414.34	92.64	163.47	-414.20	
0.00	0.00	-8.00	92.39	165.72	-415.02	92.24	165.68	-415.26	
0.00	0.00	-6.00	92.51	165.79	-415.01	92.55	165.73	-415.14	
0.00	0.00	-4.00	92.63	165.86	-415.00	92.60	165.83	-415.12	
0.00	0.00	-2.00	92.75	165.93	-414.99	92.81	165.89	-415.08	
0.00	0.00	0.00	92.87	166.00	-414.98	92.87	165.97	-415.04	
0.00	0.00	2.00	92.99	166.07	-414.97	93.06	166.02	-415.03	
0.00	0.00	4.00	93.11	166.14	-414.96	93.12	166.11	-414.95	
0.00	0.00	6.00	93.23	166.22	-414.95	93.31	166.16	-414.96	
0.00	0.00	8.00	93.35	166.30	-414.94	93.40	166.23	-414.90	

Table B5: Measurement values for $I = 250$ mA

x (cm)	y (cm)	z (cm)	B_x (mG)	Reconstruction			B_x (mG)	Measured	
				B_y (mG)	B_z (mG)	B_z (mG)		B_y (mG)	B_z (mG)
-16.00	0.00	0.00	90.90	166.23	-415.93	91.17	166.43	-415.83	
-14.00	0.00	0.00	91.14	166.18	-415.82	91.39	166.43	-415.74	
-12.00	0.00	0.00	91.38	166.14	-415.70	91.60	166.38	-415.64	
-10.00	0.00	0.00	91.62	166.11	-415.58	91.81	166.32	-415.58	
-8.00	0.00	0.00	91.87	166.08	-415.46	92.02	166.26	-415.39	
-6.00	0.00	0.00	92.12	166.06	-415.34	92.23	166.18	-415.32	
-4.00	0.00	0.00	92.37	166.03	-415.22	92.43	166.13	-415.25	
-2.00	0.00	0.00	92.63	166.01	-415.10	92.61	166.05	-415.17	
0.00	0.00	0.00	92.89	165.99	-414.98	92.87	165.99	-415.06	
2.00	0.00	0.00	93.15	165.97	-414.86	93.10	165.92	-414.97	
4.00	0.00	0.00	93.41	165.94	-414.74	93.31	165.82	-414.87	
6.00	0.00	0.00	93.67	165.92	-414.62	93.56	165.77	-414.73	
8.00	0.00	0.00	93.94	165.89	-414.50	93.80	165.71	-414.65	
10.00	0.00	0.00	94.20	165.86	-414.38	94.03	165.63	-414.58	
12.00	0.00	0.00	94.47	165.83	-414.26	94.30	165.57	-414.43	
14.00	0.00	0.00	94.73	165.79	-414.14	94.55	165.48	-414.32	
16.00	0.00	0.00	94.98	165.74	-414.03	94.80	165.40	-414.25	
0.00	-20.00	0.00	92.85	168.61	-415.59	92.93	168.68	-415.65	
0.00	-18.00	0.00	92.89	168.33	-415.55	93.05	168.43	-415.59	
0.00	-16.00	0.00	92.92	168.05	-415.50	93.06	168.17	-415.52	
0.00	-14.00	0.00	92.94	167.78	-415.45	93.02	167.89	-415.48	
0.00	-12.00	0.00	92.95	167.52	-415.39	92.99	167.63	-415.45	
0.00	-10.00	0.00	92.95	167.26	-415.33	92.98	167.36	-415.36	
0.00	-8.00	0.00	92.95	167.00	-415.26	92.99	167.09	-415.29	
0.00	-6.00	0.00	92.94	166.75	-415.19	92.95	166.84	-415.22	
0.00	-4.00	0.00	92.92	166.49	-415.12	92.94	166.57	-415.14	
0.00	-2.00	0.00	92.91	166.24	-415.05	92.90	166.30	-415.12	
0.00	0.00	0.00	92.89	165.99	-414.98	92.90	166.02	-415.01	
0.00	2.00	0.00	92.86	165.73	-414.91	92.87	165.75	-414.95	
0.00	4.00	0.00	92.84	165.48	-414.83	92.88	165.47	-414.85	
0.00	6.00	0.00	92.81	165.22	-414.76	92.85	165.13	-414.79	
0.00	8.00	0.00	92.78	164.96	-414.70	92.81	164.86	-414.74	
0.00	10.00	0.00	92.76	164.70	-414.63	92.81	164.56	-414.64	
0.00	12.00	0.00	92.73	164.43	-414.56	92.81	164.31	-414.56	
0.00	14.00	0.00	92.70	164.16	-414.50	92.76	164.03	-414.45	
0.00	16.00	0.00	92.67	163.88	-414.44	92.72	163.81	-414.36	
0.00	18.00	0.00	92.64	163.59	-414.39	92.70	163.55	-414.23	
0.00	20.00	0.00	92.60	163.30	-414.34	92.66	163.29	-414.19	
0.00	0.00	-8.00	92.40	165.71	-414.95	92.26	165.66	-415.15	
0.00	0.00	-6.00	92.52	165.78	-414.96	92.58	165.71	-415.13	
0.00	0.00	-4.00	92.64	165.85	-414.96	92.63	165.82	-415.08	
0.00	0.00	-2.00	92.77	165.92	-414.97	92.81	165.86	-415.05	
0.00	0.00	0.00	92.89	165.99	-414.98	92.87	165.99	-415.07	
0.00	0.00	2.00	93.01	166.06	-414.99	93.08	166.01	-415.00	
0.00	0.00	4.00	93.13	166.13	-414.99	93.12	166.11	-415.00	
0.00	0.00	6.00	93.25	166.21	-415.00	93.32	166.15	-414.97	
0.00	0.00	8.00	93.37	166.29	-415.01	93.38	166.22	-415.02	

Table B6: Measurement values for $I = 300$ mA

x (cm)	y (cm)	z (cm)	B_x (mG)	Reconstruction			B_x (mG)	Measured	
				B_y (mG)	B_z (mG)	B_z (mG)		B_y (mG)	B_z (mG)
-16.00	0.00	0.00	90.61	166.18	-415.96	90.89	166.41	-415.80	
-14.00	0.00	0.00	90.88	166.14	-415.84	91.12	166.38	-415.71	
-12.00	0.00	0.00	91.15	166.10	-415.72	91.36	166.30	-415.64	
-10.00	0.00	0.00	91.43	166.07	-415.60	91.58	166.28	-415.54	
-8.00	0.00	0.00	91.71	166.05	-415.48	91.83	166.22	-415.42	
-6.00	0.00	0.00	92.00	166.02	-415.36	92.08	166.15	-415.33	
-4.00	0.00	0.00	92.29	166.00	-415.24	92.32	166.07	-415.24	
-2.00	0.00	0.00	92.58	165.98	-415.12	92.59	166.01	-415.14	
0.00	0.00	0.00	92.87	165.97	-415.00	92.84	165.95	-415.05	
2.00	0.00	0.00	93.17	165.95	-414.88	93.08	165.87	-414.95	
4.00	0.00	0.00	93.47	165.93	-414.76	93.34	165.82	-414.87	
6.00	0.00	0.00	93.77	165.90	-414.64	93.62	165.76	-414.77	
8.00	0.00	0.00	94.07	165.88	-414.52	93.90	165.71	-414.66	
10.00	0.00	0.00	94.37	165.85	-414.40	94.17	165.61	-414.54	
12.00	0.00	0.00	94.66	165.82	-414.28	94.45	165.56	-414.44	
14.00	0.00	0.00	94.96	165.79	-414.16	94.74	165.48	-414.32	
16.00	0.00	0.00	95.25	165.75	-414.04	95.05	165.40	-414.23	
0.00	-20.00	0.00	92.81	168.76	-415.61	92.89	168.81	-415.65	
0.00	-18.00	0.00	92.85	168.46	-415.57	93.01	168.54	-415.62	
0.00	-16.00	0.00	92.89	168.16	-415.52	93.02	168.28	-415.54	
0.00	-14.00	0.00	92.91	167.88	-415.47	93.00	167.97	-415.48	
0.00	-12.00	0.00	92.92	167.60	-415.41	92.98	167.71	-415.46	
0.00	-10.00	0.00	92.92	167.32	-415.35	92.97	167.42	-415.38	
0.00	-8.00	0.00	92.92	167.04	-415.28	92.91	167.14	-415.33	
0.00	-6.00	0.00	92.92	166.77	-415.21	92.93	166.86	-415.24	
0.00	-4.00	0.00	92.90	166.50	-415.14	92.92	166.59	-415.18	
0.00	-2.00	0.00	92.89	166.23	-415.07	92.92	166.29	-415.16	
0.00	0.00	0.00	92.87	165.97	-415.00	92.90	165.98	-415.00	
0.00	2.00	0.00	92.85	165.70	-414.93	92.90	165.73	-414.93	
0.00	4.00	0.00	92.83	165.42	-414.85	92.86	165.40	-414.86	
0.00	6.00	0.00	92.81	165.15	-414.78	92.85	165.08	-414.79	
0.00	8.00	0.00	92.78	164.87	-414.72	92.82	164.76	-414.70	
0.00	10.00	0.00	92.76	164.60	-414.65	92.80	164.46	-414.67	
0.00	12.00	0.00	92.73	164.31	-414.58	92.77	164.20	-414.59	
0.00	14.00	0.00	92.70	164.02	-414.52	92.74	163.92	-414.48	
0.00	16.00	0.00	92.68	163.73	-414.46	92.74	163.65	-414.37	
0.00	18.00	0.00	92.65	163.43	-414.41	92.70	163.37	-414.28	
0.00	20.00	0.00	92.62	163.12	-414.36	92.67	163.10	-414.26	
0.00	0.00	-8.00	92.38	165.68	-414.89	92.24	165.63	-415.11	
0.00	0.00	-6.00	92.51	165.75	-414.92	92.52	165.68	-415.05	
0.00	0.00	-4.00	92.63	165.82	-414.95	92.59	165.80	-415.10	
0.00	0.00	-2.00	92.75	165.89	-414.97	92.80	165.86	-415.07	
0.00	0.00	0.00	92.87	165.97	-415.00	92.85	165.93	-415.06	
0.00	0.00	2.00	92.99	166.04	-415.02	93.05	165.98	-415.04	
0.00	0.00	4.00	93.11	166.11	-415.05	93.11	166.10	-415.05	
0.00	0.00	6.00	93.24	166.19	-415.08	93.32	166.13	-415.06	
0.00	0.00	8.00	93.36	166.26	-415.10	93.35	166.19	-415.08	

Table B7: Measurement values for $I = 350$ mA

x (cm)	y (cm)	z (cm)	B_x (mG)	Reconstruction			B_x (mG)	Measured	
				B_y (mG)	B_z (mG)	B_z (mG)		B_y (mG)	B_z (mG)
-16.00	0.00	0.00	90.29	166.16	-415.97	90.61	166.37	-415.86	
-14.00	0.00	0.00	90.60	166.12	-415.85	90.87	166.34	-415.72	
-12.00	0.00	0.00	90.91	166.08	-415.73	91.13	166.32	-415.62	
-10.00	0.00	0.00	91.23	166.05	-415.61	91.41	166.26	-415.52	
-8.00	0.00	0.00	91.54	166.03	-415.49	91.69	166.20	-415.41	
-6.00	0.00	0.00	91.86	166.01	-415.37	92.00	166.16	-415.33	
-4.00	0.00	0.00	92.19	165.99	-415.24	92.25	166.08	-415.24	
-2.00	0.00	0.00	92.52	165.97	-415.12	92.54	166.01	-415.17	
0.00	0.00	0.00	92.85	165.96	-415.00	92.83	165.96	-415.11	
2.00	0.00	0.00	93.18	165.94	-414.88	93.12	165.89	-414.96	
4.00	0.00	0.00	93.51	165.92	-414.76	93.42	165.81	-414.83	
6.00	0.00	0.00	93.85	165.90	-414.63	93.71	165.76	-414.75	
8.00	0.00	0.00	94.19	165.88	-414.51	93.98	165.70	-414.65	
10.00	0.00	0.00	94.52	165.86	-414.39	94.34	165.61	-414.51	
12.00	0.00	0.00	94.86	165.83	-414.27	94.65	165.55	-414.43	
14.00	0.00	0.00	95.19	165.80	-414.15	94.97	165.47	-414.28	
16.00	0.00	0.00	95.52	165.76	-414.03	95.31	165.40	-414.21	
0.00	-20.00	0.00	92.76	168.93	-415.63	92.89	168.97	-415.68	
0.00	-18.00	0.00	92.80	168.61	-415.58	92.99	168.71	-415.52	
0.00	-16.00	0.00	92.84	168.30	-415.53	92.98	168.39	-415.55	
0.00	-14.00	0.00	92.86	168.00	-415.48	92.98	168.13	-415.52	
0.00	-12.00	0.00	92.88	167.70	-415.42	92.94	167.79	-415.43	
0.00	-10.00	0.00	92.89	167.40	-415.35	92.93	167.50	-415.37	
0.00	-8.00	0.00	92.89	167.11	-415.29	92.90	167.21	-415.37	
0.00	-6.00	0.00	92.88	166.82	-415.22	92.90	166.92	-415.24	
0.00	-4.00	0.00	92.87	166.53	-415.14	92.88	166.61	-415.16	
0.00	-2.00	0.00	92.86	166.24	-415.07	92.87	166.30	-415.11	
0.00	0.00	0.00	92.85	165.96	-415.00	92.87	165.99	-415.06	
0.00	2.00	0.00	92.83	165.67	-414.93	92.86	165.70	-414.89	
0.00	4.00	0.00	92.81	165.38	-414.86	92.85	165.35	-414.85	
0.00	6.00	0.00	92.79	165.09	-414.78	92.80	165.01	-414.83	
0.00	8.00	0.00	92.77	164.80	-414.71	92.80	164.68	-414.77	
0.00	10.00	0.00	92.74	164.50	-414.65	92.77	164.37	-414.66	
0.00	12.00	0.00	92.72	164.20	-414.58	92.76	164.10	-414.55	
0.00	14.00	0.00	92.70	163.89	-414.52	92.74	163.77	-414.48	
0.00	16.00	0.00	92.67	163.58	-414.46	92.71	163.49	-414.42	
0.00	18.00	0.00	92.64	163.27	-414.41	92.69	163.20	-414.33	
0.00	20.00	0.00	92.62	162.94	-414.35	92.66	162.93	-414.22	
0.00	0.00	-8.00	92.35	165.67	-414.82	92.19	165.64	-415.04	
0.00	0.00	-6.00	92.48	165.74	-414.86	92.49	165.69	-415.00	
0.00	0.00	-4.00	92.60	165.81	-414.91	92.57	165.79	-415.07	
0.00	0.00	-2.00	92.72	165.89	-414.95	92.77	165.83	-415.05	
0.00	0.00	0.00	92.85	165.96	-415.00	92.83	165.94	-415.06	
0.00	0.00	2.00	92.97	166.03	-415.05	93.02	165.98	-415.09	
0.00	0.00	4.00	93.09	166.11	-415.09	93.09	166.07	-414.96	
0.00	0.00	6.00	93.21	166.18	-415.14	93.27	166.11	-415.14	
0.00	0.00	8.00	93.34	166.26	-415.18	93.32	166.19	-415.16	

Table B8: Caltech test measurement and COMSOL simulation values

x (cm)	B_x (mG) COMSOL	B_x (mG) Reconstruction
-16.9	33.940	33.944
-14.8	33.916	33.918
-12.6	33.895	33.896
-10.7	33.879	33.879
-8.9	33.865	33.865
-6.9	33.853	33.854
-5.0	33.844	33.844
-2.5	33.837	33.839
0.0	33.834	33.835
2.5	33.837	33.839
5.0	33.844	33.844
6.93	33.853	33.854
8.9	33.865	33.865
10.7	33.880	33.880
12.6	33.897	33.896
14.8	33.920	33.918
16.9	33.947	33.944

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